

Answers to Odd-Numbered Problems

CHAPTER 1

Exercises 1.1, page 5

1. ODE, 2nd-order, ind. var. t , dep. var. x , linear
3. ODE, 1st-order, ind. var. x , dep. var. y , nonlinear
5. ODE, 1st-order, ind. var. x , dep. var. y , nonlinear
7. ODE, 1st-order, ind. var. t , dep. var. p , nonlinear
9. ODE, 2nd-order, ind. var. x , dep. var. y , linear
11. PDE, 2nd-order, ind. var. t, r , dep. var. N
13. $dp/dt = kp$, where k is the proportionality constant
15. $dT/dt = k(M - T)$, where k is the proportionality constant
17. Kevin wins by $6\sqrt{3} - 4\sqrt{6} \approx 0.594$ sec.

Exercises 1.2, page 13

3. Yes 5. No 7. Yes 9. No 11. Yes
13. Yes 19. The left-hand side is always ≥ 4 .
21. (a) $1/3, -3$ (b) $1 \pm \sqrt{6}$ 23. Yes 25. Yes
27. No 29. (b) Yes 31. (a) No
(c) $y = 2x$ and $y = -2x$

Exercises 1.3, page 21

1. (b)

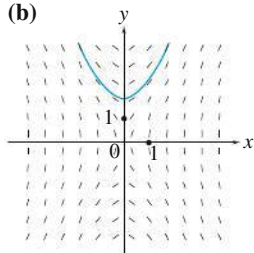


Figure B.1

Solution to Problem 1(b)

(c)

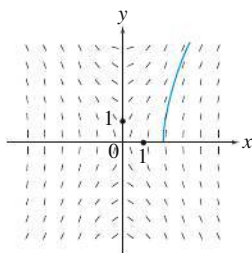


Figure B.2

Solution to Problem 1(c)

- (d) The solutions in parts (b) and (c) both become infinite and have the line $y = 2x$ as an asymptote as $x \rightarrow \infty$. As $x \rightarrow -\infty$, the solution in part (b) becomes infinite and has $y = -2x$ as an asymptote, but the solution in part (c) does not even exist for x negative.
3. All solutions have limiting value 8 as $t \rightarrow +\infty$.

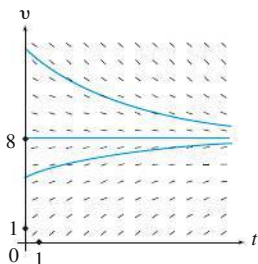


Figure B.3 Solutions to Problem 3 satisfying $v(0) = 5$, $v(0) = 8$, and $v(0) = 15$

5. (a)

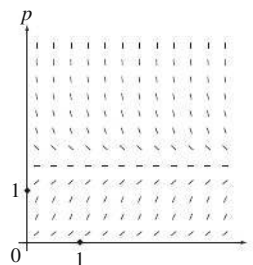


Figure B.4 Direction field for Problem 5(a)

- (b) $\lim_{t \rightarrow \infty} p(t) = 3/2$ (c) $\lim_{t \rightarrow \infty} p(t) = 3/2$
(d) No

7. (a)

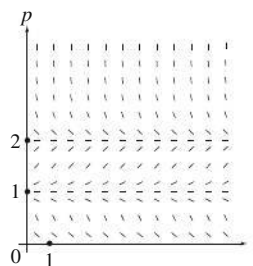


Figure B.5 Direction field for Problem 7(a)

- (b) 2 (c) 2 (d) 0 (e) No

9. (d) It increases and asymptotically approaches the line $y = x - 1$.
(f) and (g)

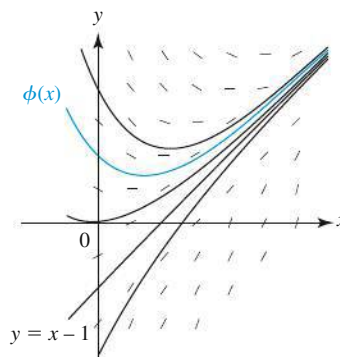


Figure B.6 Direction field and sketch of $\phi(x)$ for Problems 9 (f), (g)

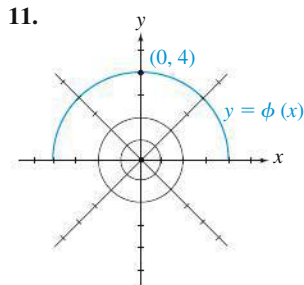


Figure B.7 Solution to Problem 11

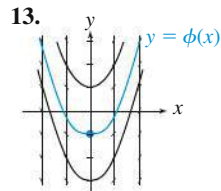


Figure B.8 Solution to Problem 13

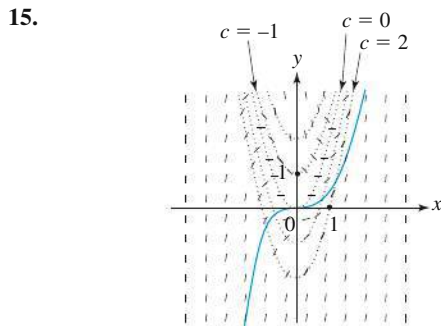


Figure B.9 Solution to Problem 15

17. It approaches 3.

Exercises 1.4, page 28

1. (rounded to three decimal places)

x_n	0.1	0.2	0.3	0.4	0.5
y_n	4.000	3.998	3.992	3.985	3.975

3. (rounded to three decimal places)

x_n	0.1	0.2	0.3	0.4	0.5
y_n	1.100	1.220	1.362	1.528	1.721

5. (rounded to three decimal places)

x_n	1.1	1.2	1.3	1.4	1.5
y_n	0.100	0.209	0.325	0.444	0.564

7. (rounded to three decimal places)

N	h	y_N	9. x_n	y_n
1	π	3.142	1.1	-0.9
2	$\pi/2$	1.571	1.2	-0.81654
4	$\pi/4$	1.207	1.3	-0.74572
8	$\pi/8$	1.148	1.4	-0.68480
			1.5	-0.63176
			1.6	-0.58511
			1.7	-0.54371
			1.8	-0.50669
			1.9	-0.47335
			2.0	-0.44314

11. $x(1)$ stabilizes at 1.56 ± 0.01 for h below 2^{-10} ; $[\tan(1) = 1.5574 \dots] x(t_0) = 1$ for some t_0 in $(0.78, 0.80)$; $[\arctan(1) = 0.785 \dots]$.
 15. (a) $T(30) = 311.7$ (b) $T(60) = 298.2$

Review Problems, page 29

1. ind. var. t , dep. var. x , nonlinear
 3. ind. var. y , dep. var. x , linear
 5. ind. var. t , dep. var. v , linear
 7. True 9. False 11. True 13. True
 15. $y_n = \left(1 - \frac{1}{n}\right)^n$, $y(1) = e^{-1} \approx 0.3679$
 17. (a) 500 (b) No (c) 500

CHAPTER 2

Exercises 2.2, page 46

1. No 3. Yes 5. Yes
 7. $y = \pm \sqrt[4]{\ln(x^4) + C}$
 9. $e^{2x}(2x - 1) + 4e^{-t}(t + 1) = C$
 11. $v = \pm \frac{\sqrt{1 - Cx^{-8/3}}}{2}$
 13. $\frac{y}{\sqrt{1 + y^2}} - x^3 = C$
 15. $y = \pm \sqrt{C \exp[\exp(-x^2)]} - 1$
 17. $y = \tan\left(-\ln|\cos x| + \frac{\pi}{3}\right)$
 19. $y = (\sin x + 1)^2 - 1$
 21. $y^2/2 + \ln y = \sin \theta - \theta \cos \theta + 1/2 - \pi$
 23. $y = \arctan(t^2 + 1)$
 25. $y = 4e^{x^3/3} - 1$
 27. (a) $y(x) = \int_0^x e^{t^2} dt$
 (b) $y(x) = \left(1 + 3 \int_0^x e^{t^2} dt\right)^{1/3}$
 (c) $y(x) = \tan\left(\int_0^x \sqrt{1 + \sin t} dt + \pi/4\right)$
 (d) $y(0.5) \approx 1.381$
 29. (d) $\partial f/\partial y$ is not continuous at $(0, 0)$.
 31. (a) $x^2 + y^{-2} = C$
 (b) $\frac{1}{\sqrt{1 - x^2}}$; $\frac{1}{\sqrt{4 - x^2}}$; $\frac{1}{\sqrt{1/4 - x^2}}$
 (c) $-1 < x < 1$; $-2 < x < 2$; $-\frac{1}{2} < x < \frac{1}{2}$
 (d) $|x| < \frac{1}{a}$ is domain.
 33. 28.1 kg
 35. (a) 82.2 min (b) 31.8 min
 (c) Never attains desired temperature

37. (a) \$1105.17 (b) 27.73 years
 (c) \$4427.59
 39. A wins.

Exercises 2.3, page 54

1. Linear 3. Both 5. Neither
 7. $y = (1/2)e^{3x} + Ce^x$
 9. $r = \sin \theta + C \cos \theta$
 11. $y = -t - 2 + Ce^t$
 13. $x = y^3 + Cy^{-2}$
 15. $y = 1 + C(x^2 + 1)^{-1/2}$
 17. $y = xe^x - x$
 19. $x = \frac{t^3}{6} \ln t - \frac{t^3}{36} + \frac{1}{2t} - \frac{17}{36t^3}$
 21. $y = x^2 \cos x - \pi^2 \cos x$
 23. $y(t) = (38/3)e^{-5t} - (8/3)e^{-20t}$
 25. (b) $y(3) \approx 0.183$
 27. (b) 0.9960 (c) 0.9486, 0.9729
 29. $x = e^{4y}/2 + Ce^{2y}$
 31. (a) $y = x - 1 + Ce^{-x}$
 (b) $y = x - 1 + 2e^{-x}$
 (c) $y = x/3 - 1/9 + Ce^{-3x}$
 (d) $y = \begin{cases} x - 1 + 2e^{-x}, & \text{if } 0 \leq x \leq 2 \\ x/3 - 1/9 + (4e^6/9 + 2e^4)e^{-3x}, & \text{if } 2 < x \end{cases}$
 (e)

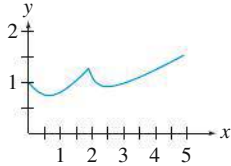


Figure B.10 Solution to Problem 31(e)

33. (a) $y = x$ is only solution in a neighborhood of $x = 0$.
 (b) $y = -3x + Cx^2$ satisfies $y(0) = 0$ for any C .
 35. (a) 0.0281 kg/L (b) 0.0598 kg/L
 37. $x(t) = \frac{1}{2} - \frac{2 \cos(\pi t/12)}{4 + (\pi/12)^2} - \frac{(\pi/12) \sin(\pi t/12)}{4 + (\pi/12)^2}$
 $+ \left(\frac{19}{2} + \frac{2}{4 + (\pi/12)^2} \right) e^{-2t}$
 39. 71.8°F at noon; 26.9°F at 5 P.M.

Exercises 2.4, page 64

1. Linear with y as dep. var.
 3. Separable
 5. Separable, also linear with x as dep. var.
 7. Exact
 9. $y = (C - 3x)/(x^2 - 1)$
 11. $e^x \sin y - x^3 + \sqrt[3]{y} = C$
 13. $y = \frac{(t-1)e^t + C}{1 + e^t}$
 15. $r = (C - e^\theta) \sec \theta$
 17. Not exact
 19. $x^2 - y^2 + \arctan(xy) = C$

21. $\ln x + x^2 y^2 - \sin y = \pi^2$
 23. $y = -2/(te^t + 2)$
 25. $\sin x - x \cos x = \ln y + 1/y + \pi - 1$ (equation is separable, not exact)
 27. (a) $-\ln|y| + f(x)$
 (b) $\cos x \sin y - y^2/2 + f(x)$, where f is a function of x only
 29. (c) $y = x^2/(C - x)$ (d) Yes, $y \equiv 0$
 33. (a) $x = cy^2$; $x \equiv 0$; $y \equiv 0$
 (b) $x^2 + 4y^2 = c$
 (c) $2y^2 \ln y - y^2 + 2x^2 = c$
 (d) $2x^2 + y^2 = c$

Exercises 2.5, page 69

1. Integrating factor depending on x alone
 3. Exact
 5. Linear with x as dep. var., has integrating factor depending on x alone
 7. $\mu = y^{-4}$; $x^2 y^{-3} - y^{-1} = C$ and $y \equiv 0$
 9. $\mu = x^{-2}$; $y = x^4/3 - x \ln|x| + Cx$ and $x \equiv 0$
 11. $\mu = y^{-2}$; $x^2 y^{-1} + x = C$ and $y \equiv 0$
 13. $\mu = xy$; $x^2 y^3 - 2x^3 y^2 = C$
 15. (a) $\mu(z) = \exp[\int H(z) dz]$; $z = xy$
 (b) $\ln(x^3 y) + 2xy = 2$
 17. (a) $\frac{\partial N/\partial x - \partial M/\partial y}{x^2 M - 2xyN}$ depends only on $x^2 y$.
 (b) $\exp(x^2 y)[x^2 + 2xy] = C$
 19. (b) $\mu = e^y$; $x = y - 1 + Ce^{-y}$

Exercises 2.6, page 76

1. Homogeneous and Bernoulli 3. Bernoulli
 5. Homogeneous and Bernoulli
 7. $y' = G(ax + by)$
 9. $y = -x/(\ln|x| + C)$ and $y \equiv 0$
 11. $y = x/(\ln|x| + C)$ and $y \equiv 0, x \equiv 0$
 13. $\sqrt{1 + x^2/t^2} = \ln|t| + C$
 15. $(x^2 - 4y^2)^3 x^2 = C$
 17. $y = (x + C)^2/4 - x$ and $y = -x$
 19. $y = x + (6 + 4Ce^{2x})/(1 + Ce^{2x})$ and $y = x + 4$
 21. $y = 2/(Cx - x^3)$ and $y \equiv 0$
 23. $y = 5x^2/(x^5 + C)$ and $y \equiv 0$
 25. $x^{-2} = 2t^2 \ln|t| + Ct^2$ and $x \equiv 0$
 27. $r = \theta^2/(C - \theta)$ and $r \equiv 0$
 29. $2 \arctan\left(\frac{y-3}{x+2}\right) - \ln[(x+2)^2 + (y-3)^2] = C$
 31. $(2x + 2y - 3)^3 = C(2x + y - 2)^2$
 33. $x^2 + t^2 - Ct = 0$ and $t \equiv 0$
 35. $y = -3/(x^4 + Cx)$ and $y \equiv 0$
 37. $y = (\theta/4)(\ln|\theta| + C)^2$ and $\theta \equiv 0$
 39. $\sin(x + y) + \cos(x + y) = Ce^{x-y}$
 41. $(x - y + 2)^2 = Ce^{2x} + 1$
 45. $(y - 4x)^2(y + x)^3 = C$
 47. (a) $v' + [2Pu + Q]v = -P$
 (b) $y = x + 5x/(C - x^5)$

Review Problems, page 79

1. $e^x + ye^{-y} = C$
3. $x^2y - x^3 + y^{-2} = C$
5. $y + x \sin(xy) = C$
7. $t = C \exp(1/(7y^7))$
9. $(x^2 + 4y^2)^3 x^2 = C$
11. $\tan(t - x) + t = C$
13. $y = -(x^2/2)\cos(2x) + (x/4)\sin(2x) + Cx$
15. $y = 2x + 3 - (x + C)^2/4$
17. $y = 2/(1 + Ce^{2\theta})$ and $y \equiv 0$
19. $y^2 = x^2 + Cx^3$ and $x \equiv 0$
21. $xy - x^2 - x + y^2/2 - 4y = C$
23. $y^2 + 2xy - x^2 = C$
25. $x^2y^{-2} - 2xy^{-1} - 4xy^{-2} = C$ and $y \equiv 0$
27. $[(y-4)^2 - 3(x-3)^2] \left| \frac{\sqrt{3}(x-3) + (y-4)}{\sqrt{3}(x-3) - (y-4)} \right|^{1/\sqrt{3}} = C$
29. $x^4y^3 - 3x^3y^2 + x^4y^2 = C$
31. $y = -x^3/2 + 7x/2$
33. $x = -t - 2 + 3e^{-t}$
35. $y = -2x\sqrt{2x^2 - 1}$
37. $\ln[(y-2)^2 + 2(x-1)^2] + \sqrt{2} \arctan\left[\frac{y-2}{\sqrt{2}(x-1)}\right] = \ln 2$
39. $y = \sqrt{(19x^4 - 1)/2}$
41. $y(t) = e^{-t} \int_2^t \frac{e^r}{1+r^2} dr + 3e^{-(t-2)}$, $y(3) \approx 1.1883 \dots$

CHAPTER 3

Exercises 3.2, page 100

1. $5 - 4.5e^{-2t/25}$ kg; 5.07 min
3. $(0.4)(100 - t) - (3.9 \times 10^{-7})(100 - t)^4$ L; 19.96 min
5. 0.0097%; 73.24 h
7. 20 min later; $1/e$ times as salty
9. 110,868
13. 5970; 6000
15. 6572; 6693

17. (a)

Year	$\frac{1}{p} \frac{dp}{dt}$	Logistic (Least Squares)
1790	0.0351	3.29
1800	0.0363	4.51
1810	0.0331	6.16
1820	0.0335	8.41
1830	0.0326	11.45
1840	0.0359	15.53
1850	0.0356	20.97
1860	0.0267	28.14
1870	0.0260	37.45
1880	0.0255	49.31
1890	0.0210	64.07

Year	$\frac{1}{p} \frac{dp}{dt}$	Logistic (Least Squares)
1900	0.0210	81.89
1910	0.0150	102.65
1920	0.0162	125.87
1930	0.0073	150.64
1940	0.0145	175.80
1950	0.0185	200.10
1960	0.0134	222.47
1970	0.0114	242.16
1980	0.0098	258.81
1990	0.0132	272.44
2000	0.0097	283.28
2010		291.73

- (b) $p_1 = 316.920$, $A = 0.00010050$
- (c) $p_0 = 3.28780$ (using all data, including 2010)
- (d) See table in part (a).
19. $(1/2)\ln(15) \approx 1.354$ yr; 14 million tons per yr
21. 1 hr; 2 h
23. 11.7%
25. 31,606 yr
27. e^{-2t} kg of Hh, $2e^{-t} - 2e^{-2t}$ kg of It, and $1 - 2e^{-t} + e^{-2t}$ kg of Bu

Exercises 3.3, page 107

1. 20.7 min
3. 22.6 min
5. 9:08 A.M.
7. 28.3°C; 32.5°C; 1:16 P.M.
9. 16.3°C; 19.1°C; 31.7°C; 28.9°C
11. 39.5 min
13. 148.6°F
15. $T - M = C(T + M)\exp[2 \arctan(T/M) - 4M^3kt]$; for T near M , $M^4 - T^4 \approx 4M^3(M - T)$, and so $dT/dt \approx k_1(M - T)$, where $k_1 = 4M^3k$

Exercises 3.4, page 115

1. $(0.981)t + (0.0981)e^{-10t} - 0.0981$ m; 1019 sec
3. 18.6 sec
5. $4.91t + 22.55 - 22.55e^{-2t}$ m; 97.3 sec
7. 241 sec
9. $95.65t + 956.5e^{-t/10} - 956.5$ m; 13.2 sec
11. $e^{bv}|bv - mg|^{mg} = e^{v_0b}|bv_0 - mg|^{mg}e^{-b^2x/m}$
13. 2.69 sec; 101.19 m
15. $(\omega_0 - T/k)e^{-kt/I} + T/k$
17. 300 sec
19. $2636e^{-t/20} + 131.8t - 2636$ m; 1.768 sec
21. $5e^{-2t}/2 + 6t - 5/2$; 6 m/sec
23. Sailboat B
25. (e) 11.18 km/sec
- (f) 2.38 km/sec

Exercises 3.5, page 121

1. $I = \{1.44e^{-100t} + \cos 120t + 1.2 \sin 120t\}/2.44$;
 $E_L = (-7.2e^{-100t} - 6 \sin 120t + 7.2 \cos 120t)/2.44$
3. $-(\ln .4) \times 10^{-10} \approx 9.2 \times 10^{-11}$ sec
5. $VI = IRI = I^2R$; $VI = L \frac{dI}{dt} I = \frac{d}{dt} \frac{LI^2}{2}$;
 $VI = E_C \frac{dCE_C}{dt} = \frac{d}{dt} \frac{CE_C^2}{2}$
7. $-(10 \ln .1)/3 \approx 7.68$ sec

Exercises 3.6, page 129

3. h	" e "	7. x_n	y_n
1	3	1.1	0.10450
0.1	2.72055	1.2	0.21668
0.01	2.71830	1.3	0.33382
0.001	2.71828	1.4	0.45300
0.0001	2.71828	1.5	0.57135

9. x_n	y_n
0.2	0.61784
0.4	1.23864
0.6	1.73653
0.8	1.98111
1.0	1.99705
1.2	1.88461
1.4	1.72447
1.6	1.56184
1.8	1.41732
2.0	1.29779

11. $\phi(1) \approx x(1; 2^{-3}) = 1.25494$

13. $\phi(1) \approx y(1; 2^{-3}) = 0.71698$

15. $x = 1.27$

17. x_n	$y_n(h = 0.2)$	$y_n(h = 0.1)$	$y_n(h = 0.025)$
0.1		-1	0.06250
0.2	-3	1	0.00391
0.3		-1	0.00024
0.4	9	1	0.00002
0.5		-1	0.00000
0.6	-27	1	0.00000
0.7		-1	0.00000
0.8	81	1	0.00000
0.9		-1	0.00000
1.0	-243	1	0.00000

We conclude that step size can dramatically affect convergence.

19. T_n			
Time	$K = 0.2$	$K = 0.4$	$K = 0.6$
Midnight	65.0000	65.0000	65.0000
4 A.M.	69.1639	68.5644	68.1300
8 A.M.	71.4836	72.6669	73.6678
Noon	72.9089	75.1605	76.9783
4 P.M.	72.0714	73.5977	74.7853
8 P.M.	69.8095	69.5425	69.2831
Midnight	68.3852	67.0500	65.9740

Exercises 3.7, page 139

$$1. y_{n+1} = y_n + h \cos(x_n + y_n) - \frac{h^2}{2} \sin(x_n + y_n) [1 + \cos(x_n + y_n)]$$

$$3. y_{n+1} = y_n + h(x_n - y_n) + \frac{h^2}{2}(1 - x_n + y_n) - \frac{h^3}{6}(1 - x_n + y_n) + \frac{h^4}{24}(1 - x_n + y_n)$$

5. Order 2, $\phi(1) \approx 1.3725$; order 4, $\phi(1) \approx 1.3679$

7. -11.7679 9. 1.36789 11. $x = 1.41$

13. $x = 0.50$

15. x_n	y_n
0.5	0.21462
1.0	0.13890
1.5	-0.02668
2.0	-0.81879
2.5	-1.69491
3.0	-2.99510

19. $v(3) \approx 0.24193$ with $h = 0.0625$

21. $z(1) \approx 2.87083$ with $h = 0.03125$

CHAPTER 4

Exercises 4.1, page 156

3. Both approach zero. 5. 0

7. $y(t) = -(30/61) \cos 3t - (25/61) \sin 3t$

9. $y(t) = -2 \cos 2t + (3/2) \sin 2t$

Exercises 4.2, page 164

1. $c_1 e^{t/2} + c_2 e^{-4t}$

3. $c_1 e^{-3t} + c_2 e^{-2t}$

5. $c_1 e^{-4t} + c_2 t e^{-4t}$

7. $c_1 e^{t/2} + c_2 e^{-2t/3}$

9. $c_1 e^{t/2} + c_2 t e^{t/2}$

11. $c_1 e^{-5t/2} + c_2 t e^{-5t/2}$

13. $3e^{-4t}$ 15. $\frac{4}{3}e^t - \frac{1}{3}e^{3t}$

17. $\left(\frac{7t}{3} + 2\right)e^{3t}$ 19. $e^{-t} - 2te^{-t}$

21. (a) $ar + b = 0$ (b) $ce^{-bt/a}$

23. $ce^{-4t/5}$ 25. $ce^{13t/6}$

27. Lin. dep. 29. Lin. indep. 31. Lin. dep.

33. If $c_1 \neq 0$, then $y_1 = -(c_2/c_1)y_2$.

35. (a) Lin. indep. (b) Lin. dep. (c) Lin. indep. (d) Lin. dep.

37. $c_1 e^t + c_2 e^{(-1-\sqrt{5})t} + c_3 e^{(-1+\sqrt{5})t}$

39. $c_1 e^{-2t} + c_2 t e^{-2t} + c_3 e^{2t}$

41. $c_1 e^{-3t} + c_2 e^{-2t} + c_3 e^{2t}$

43. $3 + e^t - 2e^{-t}$

45. (a) $c_1 e^{r_1 t} + c_2 e^{r_2 t} + c_3 e^{r_3 t}$ (where $r_1 = -4.832$, $r_2 = -1.869$, and $r_3 = 0.701$)

(b) $c_1 e^{r_1 t} + c_2 e^{-r_1 t} + c_3 e^{r_2 t} + c_4 e^{-r_2 t}$

(where $r_1 = 1.176$, $r_2 = 1.902$)

(c) $c_1 e^{-t} + c_2 e^t + c_3 e^{-2t} + c_4 e^{2t} + c_5 e^{3t}$

Exercises 4.3, page 172

1. $c_1 \cos 3t + c_2 \sin 3t$

3. $c_1 e^{3t} \cos t + c_2 e^{3t} \sin t$

5. $c_1 e^{-2t} \cos \sqrt{2}t + c_2 e^{-2t} \sin \sqrt{2}t$

7. $c_1 e^{-t/2} \cos(\sqrt{5}t/2) + c_2 e^{-t/2} \sin(\sqrt{5}t/2)$

9. $c_1 e^t + c_2 e^{t^2}$

11. $c_1 e^{-5t} + c_2 t e^{-5t}$
 13. $c_1 e^t \cos 5t + c_2 e^t \sin 5t$
 15. $c_1 e^{(3+\sqrt{53})t/2} + c_2 e^{(3-\sqrt{53})t/2}$
 17. $c_1 e^{t/2} \cos(3\sqrt{3}t/2) + c_2 e^{t/2} \sin(3\sqrt{3}t/2)$
 19. $c_1 e^t + c_2 e^{-t} \cos 2t + c_3 e^{-t} \sin 2t$
 21. $2e^{-t} \cos t + 3e^{-t} \sin t$
 23. $(\sqrt{2}/4)[e^{(2+\sqrt{2})t} - e^{(2-\sqrt{2})t}]$
 25. $e^t \sin t - e^t \cos t$
 27. $e^{2t} - \sqrt{2} e^t \sin \sqrt{2}t$
 29. (a) $c_1 e^{-t} + c_2 e^t \cos \sqrt{2}t + c_3 e^t \sin \sqrt{2}t$
 (b) $c_1 e^{2t} + c_2 e^{-2t} \cos 3t + c_3 e^{-2t} \sin 3t$
 (c) $c_1 \cos 2t + c_2 \sin 2t + c_3 \cos 3t + c_4 \sin 3t$
 31. (a) Oscillatory (b) Tends to zero
 (c) Tends to $-\infty$ (d) Tends to $-\infty$
 (e) Tends to $+\infty$
 33. (a) $y(t) = 0.3e^{-3t} \cos 4t + 0.2e^{-3t} \sin 4t$
 (b) $[\pi + \arctan(-1.5)]/4$
 (c) $2/\pi$
 (d) Decreases the frequency of oscillation, introduces the factor e^{-3t} , causing the solution to decay to zero
 35. $b \geq 2\sqrt{Ik}$
 37. (a) $c_1 \cos t + c_2 \sin t + c_3 t \cos t + c_4 t \sin t$
 (b) $(c_1 + c_2)e^{-t} \cos(\sqrt{3}t) + (c_3 + c_4t)e^{-t} \sin(\sqrt{3}t)$

Exercises 4.4, page 180

1. No 3. Yes 5. No 7. Yes
 9. $y_p \equiv -3$
 11. $y_p(x) = [(\ln 2)^2 + 1]^{-1} 2^x$
 13. $\cos 3t$
 15. $xe^x/2 + 3e^x/4$
 17. $-2t \cos 2t$
 19. $\left(\frac{t}{13} + \frac{8}{169}\right)te^{-3t}$ 21. $t^3 e^{2t}/6$
 23. $-\frac{1}{21}\theta^3 - \frac{1}{49}\theta^2 - \frac{2}{343}\theta$
 25. $e^{2t}(\cos 3t + 6 \sin 3t)$
 27. $(A_3 t^4 + A_2 t^3 + A_1 t^2 + A_0 t) \cos 3t + (B_3 t^4 + B_2 t^3 + B_1 t^2 + B_0 t) \sin 3t$
 29. $e^t(A_4 t^4 + A_3 t^3 + A_2 t^2 + A_1 t + A_0)$
 31. $(A_3 t^4 + A_2 t^3 + A_1 t^2 + A_0 t)e^{-t} \cos t + (B_3 t^4 + B_2 t^3 + B_1 t^2 + B_0 t)e^{-t} \sin t$
 33. $(1/5) \cos t + (2/5) \sin t$
 35. $\left(\frac{1}{10}t^2 - \frac{4}{25}t\right)e^t$

Exercises 4.5, page 185

1. (a) $5 \cos t$ (b) $\cos t - e^{2t}$ (c) $4 \cos t + 6e^{2t}$
 3. $y = -t + c_1 e^t + c_2 e^{-t}$
 5. $\theta = t - 1 + c_1 e^{2t} + c_2 e^{-t}$
 7. $y = \tan x + c_1 e^{\sqrt{2}x} + c_2 e^{-\sqrt{2}x}$
 9. Yes 11. No 13. Yes 15. No
 17. $y(t) = -t^2 + \frac{4t}{3} + \frac{1}{9} + c_1 e^{3t} + c_2 e^{-t}$

19. $y = (\cos x - \sin x)e^x/2 + c_1 e^x + c_2 e^{2x}$
 21. $y = (1/2)\theta e^{-\theta} \sin \theta + (c_1 \cos \theta + c_2 \sin \theta)e^{-\theta}$
 23. $y = e^t - 1$
 25. $z = e^{-x} - \cos x + \sin x$
 27. $y = -(3/10) \cos x - (1/10) \sin x - (1/20) \cos 2x + (3/20) \sin 2x$
 29. $y = -(1/2) \sin \theta - (1/3)e^{2\theta} + (3/4)e^\theta + (7/12)e^{-\theta}$
 31. $y_p = (A_1 t + A_0)t \cos t + (B_1 t + B_0)t \sin t + C \cdot 10^t$
 33. $x_p(t) = (A \cos t + B \sin t)e^t + C_2 t^2 + C_1 t + C_0 + D_1 \cos 3t + D_2 \sin 3t + E_1 \cos t + E_2 \sin t$
 35. $y_p = (A_1 t + A_0) \cos 3t + (B_1 t + B_0) \sin 3t + C e^{5t}$
 37. $y_p = t^2 + 3t - 1$
 39. $y_p = (t/10 - 4/25)te^t - 1/2$
 41. (a) $y_1 = -(2 \cos 2t + \sin 2t)e^{-t} + 2$ for $0 \leq t \leq 3\pi/2$
 (b) $y_2 = y_h = (c_1 \cos 2t + c_2 \sin 2t)e^{-t}$ for $t > 3\pi/2$
 (c) $c_1 = -2(e^{3\pi/2} + 1)$, $c_2 = -(e^{3\pi/2} + 1)$
 43. $y = -\cos t + (1/2) \sin t - (1/2)e^{-3t} + 2e^{-t}$
 45. (a) $y(t) = \frac{2V \cos(\pi/2V)}{V^2 - 1} \sin t$ for $V \neq 1$;
 $y(t) = \frac{\pi}{2} \sin t$ for $V = 1$
 (b) $V \approx 0.73$
 47. (a) $2 \sin 3t - \cos 6t$ (b) No solution
 (c) $c \sin 3t - \cos 6t$, where c is any constant

Exercises 4.6, page 191

1. $c_1 \cos 2t + c_2 \sin 2t - (1/4)(\cos 2t) \ln |\sec 2t + \tan 2t|$
 3. $c_1 e^t + c_2 t e^t + t e^t \ln |t|$
 5. $c_1 \cos 4\theta + c_2 \sin 4\theta + (\theta/4) \sin 4\theta + (1/16)(\cos 4\theta) \ln |\cos 4\theta|$
 7. $(2 \ln t - 3)t^2 e^{-2t}/4 + c_1 e^{-2t} + c_2 t e^{-2t}$
 9. $y_p = -2t - 4$
 11. $y = -(\cos t) \ln |\sec t + \tan t| + (1/10)e^{3t} - 1 + c_1 \cos t + c_2 \sin t$
 13. $c_1 \cos 2t + c_2 \sin 2t + (1/24) \sec^2 2t - 1/8 + (1/8)(\sin 2t) \ln |\sec 2t + \tan 2t|$
 15. $c_1 \cos t + c_2 \sin t - t^2 + 3 + 3t \sin t + 3(\cos t) \ln |\cos t|$
 17. $c_1 \cos 2t + c_2 \sin 2t - e^t/5 - (1/2)(\cos 2t) \ln |\sec 2t + \tan 2t|$
 19. $y = e^{1-t} - e^{t-1} + \frac{e^t}{2} \int_1^t \frac{e^{-u}}{u} du - \frac{e^{-t}}{2} \int_1^t \frac{e^u}{u} du$
 $[y(2) \approx -1.93]$
 21. 0.3785
 23. $c_1 e^t + c_2(t+1) - t^2$
 25. $c_1(5t-1) + c_2 e^{-5t} - t^2 e^{-5t}/10$

Exercises 4.7, page 199

1. Unique solution on $(-\pi/2, \pi/2)$
 3. Unique solution on $(0, \infty)$
 5. Does not apply; $t = 0$ is a point of discontinuity

7. Does not apply; not an initial value problem
9. $c_1 t + c_2 t^{-7}$
11. $c_1 t^{-2} + c_2 t^{-2} \ln t$
13. $c_1 t^{-1/3} + c_2 t^{-1/3} \ln t$
15. $c_1 t \cos [2 \ln (-t)] + c_2 t \sin [2 \ln (-t)]$
17. $t^{-4} \{ c_1 \cos [\ln (-t)] + c_2 \sin [\ln (-t)] \}$
19. $t - 3t^4$
21. $c_1(t-2) + c_2(t-2)^7$
23. (c) $t^\alpha \cos(\beta \ln |t|)$, $t^\alpha \sin(\beta \ln |t|)$; t' , $t'' \ln |t|$
25. (a) True (b) False
27. (e) No, because the coefficient of y'' vanishes at $t = 0$ and the equation cannot be written in standard form.
29. Otherwise their Wronskian would be zero at t_0 , contradicting linear independence.
31. (a) Yes (b) No (c) Yes (d) Yes
33. Cte^{-t}
35. $1 + 2t - t^2$
37. $c_1 \cos(3 \ln t) + c_2 \sin(3 \ln t) + (1/9) \cos(3 \ln t) \ln |\sec(3 \ln t) + \tan(3 \ln t)|$
39. $c_1 t + c_2 t \ln t + (1/2)t(\ln t)^2 + 3t(\ln t)[\ln |\ln t|]$
41. t^4
43. $t + 1$
45. $(t-1)e^{2t}/2$
47. (a) $(1-2t^2) \int (1-2t^2)^{-2} e^{t^2} dt$
 (b) $(3t-2t^3) \int (3t-2t^3)^{-2} e^{t^2} dt$
49. $tw'' + 2tw' + (t+1)w = 0$
51. (a) $\phi'(t_0) = \lim_{n \rightarrow \infty} \frac{\phi(t_n) - \phi(t_0)}{t_n - t_0} = \lim_{n \rightarrow \infty} \frac{0-0}{t_n - t_0} = 0$

Exercises 4.8, page 210

1. Let $Y(t) = y(-t)$. Then $Y'(t) = -y'(-t)$, $Y''(t) = y''(-t)$. But $y''(s) - sy(s) = 0$, so $y''(-t) + ty(-t) = 0$ or $Y''(t) + tY(t) = 0$.
3. The spring stiffness is $(-6y)$, so it opposes negative displacements ($y < 0$) and reinforces positive displacements ($y > 0$). Initially $y < 0$ and $y' < 0$, so the (positive) stiffness reverses the negative velocity and restores y to 0. Thereafter, $y > 0$ and the negative stiffness drives y to $+\infty$.
5. (a) $y'' = 2y^3 = \frac{d}{dy}(y^4/2)$. Thus, by setting $K = 0$ and choosing the $(-)$ sign in equation (11), we get $t = -\int \frac{dy}{\sqrt{2y^4/2}} + c = \frac{1}{y} + c$, or $y = 1/(t-c)$.
- (b) Linear dependence would imply $\frac{y_1(t)}{y_2(t)} = \frac{1/(t-c_1)}{1/(t-c_2)} = \frac{t-c_2}{t-c_1} \equiv \text{constant}$ in a neighborhood of 0, which is false if $c_1 \neq c_2$.
- (c) If $y(t) = 1/(t-c)$, then $y(0) = -1/c$, $y'(0) = -1/c^2 = -y(0)^2$, which is false for the given data.

7. (a) The velocity, which is always perpendicular to the lever arm, is $\ell d\theta/dt$. Thus, (lever arm) times (perpendicular momentum) $= \ell m \ell d\theta/dt = m \ell^2 d\theta/dt$.
- (b) The component of the gravitational force perpendicular to lever arm is $mg \sin \theta$, and is directed toward decreasing θ . Thus, torque $= -\ell mg \sin \theta$.
- (c) Torque $= \frac{d}{dt}$ (angular momentum) or $-\ell mg \sin \theta = (m \ell^2 \theta')' = m \ell^2 \theta''$.
9. 2 or -2
11. The sign of the damping coefficient $(y')^2 - 1$ indicates that low velocities are boosted by negative damping but that high velocities are slowed. Hence, one expects a limit cycle.
13. (a) Airy (b) Duffing (c) van der Pol
15. (a) Yes ($t^2 =$ positive stiffness)
 (b) No ($-t^2 =$ negative stiffness)
 (c) Yes ($y^4 =$ positive stiffness)
 (d) No ($y^5 =$ negative stiffness for $y < 0$)
 (e) Yes ($4 + 2 \cos t =$ positive stiffness)
 (f) Yes (positive stiffness and damping)
 (g) No (negative stiffness and damping)
17. $1/4\sqrt{2}$

Exercises 4.9, page 220

1. $y(t) = -(1/4) \cos 5t - (1/5) \sin 5t$
 amplitude $= \sqrt{41}/20$; period $= 2\pi/5$;
 frequency $= 5/2\pi$; $[\pi - \arctan(5/4)]/5$ sec
3. $b = 0$: $y(t) = \cos 4t$

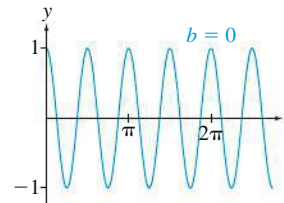


Figure B.11 $b = 0$

$$b = 6: y(t) = e^{-3t} \cos \sqrt{7}t + (3/\sqrt{7})e^{-3t} \sin \sqrt{7}t \\ = (4/\sqrt{7})e^{-3t} \sin(\sqrt{7}t + \phi), \text{ where } \phi = \arctan \sqrt{7}/3 \approx 0.723$$

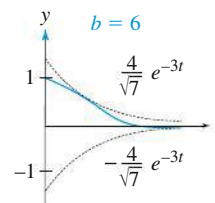


Figure B.12 $b = 6$

$$b = 8: y(t) = (1 + 4t)e^{-4t}$$

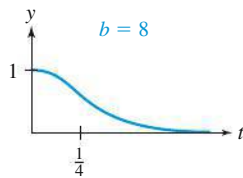


Figure B.13 $b = 8$

$$b = 10: y(t) = (4/3)e^{-2t} - (1/3)e^{-8t}$$

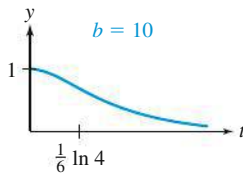


Figure B.14 $b = 10$

$$5. k = 20: y(t) = [(1 + \sqrt{5})/2]e^{(-5 + \sqrt{5})t} + [(1 - \sqrt{5})/2]e^{(-5 - \sqrt{5})t}$$

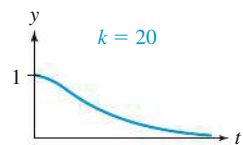


Figure B.15 $k = 20$

$$k = 25: y(t) = (1 + 5t)e^{-5t}$$

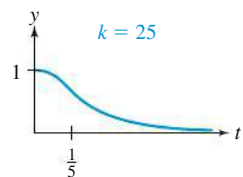


Figure B.16 $k = 25$

$$k = 30: y(t) = e^{-5t} \cos \sqrt{5}t + \sqrt{5}e^{-5t} \sin \sqrt{5}t \\ = \sqrt{6}e^{-5t} \sin(\sqrt{5}t + \phi), \text{ where } \phi = \arctan(1/\sqrt{5}) \approx 0.421$$

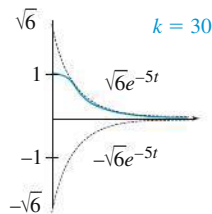


Figure B.17 $k = 30$

$$7. y(t) = (-3/4)e^{-8t} \cos 8t - e^{-8t} \sin 8t \\ = (5/4)e^{-8t} \sin(8t + \phi), \text{ where } \phi = \pi + \arctan(3/4) \approx 3.785; \\ \text{damp. factor} = (5/4)e^{-8t}; \\ \text{quasiperiod} = \pi/4; \text{quasifreq.} = 4/\pi$$

$$9. 0.242 \text{ m}$$

$$11. (10/\sqrt{9999}) \arctan(\sqrt{9999}) \approx 0.156 \text{ sec}$$

$$13. \text{Relative extrema at}$$

$$t = [\pi/3 + n\pi - \arctan(\sqrt{3}/2)]/(2\sqrt{3}) \\ \text{for } n = 0, 1, 2, \dots; \text{ but touches curves } \pm \sqrt{7/12}e^{-2t} \\ \text{at } t = [\pi/2 + m\pi - \arctan(\sqrt{3}/2)]/(2\sqrt{3}) \\ \text{for } m = 0, 1, 2, \dots$$

$$15. \text{First measure half the quasiperiod } P \text{ as the time between} \\ \text{two successive zero crossings. Then compute the ratio} \\ y(t+P)/y(t) = e^{-(b/2m)P}.$$

Exercises 4.10, page 227

$$1. M(\gamma) = 1/\sqrt{(1 - 4\gamma^2)^2 + 4\gamma^2}$$

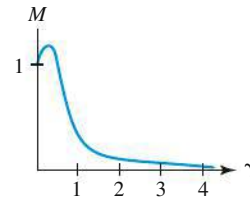


Figure B.18

$$3. y(t) = \cos 3t + (1/3)t \sin 3t$$

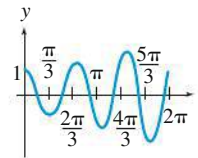


Figure B.19

$$5. (a) y(t) = -[F_0/(k - m\gamma^2)] \cos(\sqrt{k/m}t) \\ + [F_0/(k - m\gamma^2)] \cos \gamma t \\ = (F_0/[m(\omega^2 - \gamma^2)])(\cos \gamma t - \cos \omega t)$$

$$(c) y(t) = \sin 8t \sin t$$

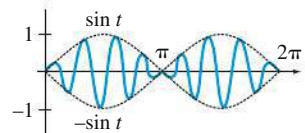


Figure B.20

$$7. y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t} + \frac{F_0 \sin(\gamma t + \theta)}{\sqrt{(k - m\gamma^2)^2 + b^2 \gamma^2}},$$

$$\text{where } r_1, r_2 = -(b/2m) \pm (1/2m)\sqrt{b^2 - 4mk} \\ \text{and } \tan \theta = (k - m\gamma^2)/(b\gamma) \text{ as in equation (7)}$$

$$9. y_p(t) = (0.08) \cos 2t + (0.06) \sin 2t \\ = (0.1) \sin(2t + \theta), \text{ where } \theta = \arctan(4/3) \approx 0.927$$

11. $y(t) = -(18/85)e^{-2t}\cos 6t - (22/255)e^{-2t}\sin 6t$
 $+ (2/\sqrt{85})\sin(2t + \theta)$, where
 $\theta = \arctan(9/2) \approx 1.352$;
 res. freq. $= 2\sqrt{2}/\pi$ cycles/sec
 13. $y_p(t) = (3/185)(8\sin 4t - 11\cos 4t)$
 15. Amp $= \sqrt{\left(-\frac{11}{986}\right)^2 + \left(-\frac{3}{1972}\right)^2} \approx 0.01$ (m),
 freq. $= 2/\pi$

Review Problems, page 231

1. $c_1e^{-9t} + c_2e^t$
 3. $c_1e^{t/2}\cos(3t/2) + c_2e^{t/2}\sin(3t/2)$
 5. $c_1e^{3t/2} + c_2e^{t/3}$
 7. $c_1e^{-t/3}\cos(t/6) + c_2e^{-t/3}\sin(t/6)$
 9. $c_1e^{7t/4} + c_2te^{7t/4}$
 11. $t^{1/2}\{c_1\cos[(\sqrt{19}/2)\ln t] + c_2\sin[(\sqrt{19}/2)\ln t]\}$
 13. $c_1\cos 4t + c_2\sin 4t + (1/17)te^t - (2/289)e^t$
 15. $c_1e^{-2t} + c_2e^{-t} + c_3e^{-t/3}$
 17. $c_1e^t + c_2e^{-t/2}\cos(\sqrt{43}t/2) + c_3e^{-t/2}\sin(\sqrt{43}t/2)$
 19. $c_1e^{-3t} + c_2e^{t/2} + c_3te^{t/2}$
 21. $c_1e^{3t/2}\cos(\sqrt{19}t/2) + c_2e^{3t/2}\sin(\sqrt{19}t/2) - e^t/5$
 $+ t^2 + 6t/7 + 4/49$
 23. $c_1\cos 4\theta + c_2\sin 4\theta$
 $- (1/16)(\cos 4\theta)\ln|\sec 4\theta + \tan 4\theta|$
 25. $c_1e^{3t/2} + c_2te^{3t/2} + e^{3t}/9 + e^{5t}/49$
 27. $c_1x + c_2x^{-2} - 2x^{-2}\ln x + x\ln x$
 29. $e^{-2t}\cos(\sqrt{3}t)$
 31. $2e^t\cos 3t - (7/3)e^t\sin 3t - \sin 3t$
 33. $-e^{-t} - 3e^{5t} + e^{8t}$
 35. $\cos \theta + 2\sin \theta + \theta\sin \theta + (\cos \theta)\ln|\cos \theta|$
 37. (a), (c), (e), and (f) have all solutions bounded as
 $t \rightarrow +\infty$
 39. $y_p(t) = (1/4)\sin 8t$; $\sqrt{62}/2\pi$

CHAPTER 5

Exercises 5.2, page 249

1. (a) $-t^3 + 3t^2 + 8$
 (b) $-2t^3 + 3t^2 + 6t + 16$
 (c) $2t^3 + 3t^2 - 16$
 (d) $-2t^3 + 3t^2 + 6t + 16$
 (e) $-2t^3 + 3t^2 + 6t + 16$
 3. $x = c_1 + c_2e^{-2t}$; $y = c_2e^{-2t}$
 5. $x \equiv -5$; $y \equiv 1$
 7. $u = c_1 - (1/2)c_2e^{-t} + (1/2)e^t + (5/3)t$;
 $v = c_1 + c_2e^{-t} + (5/3)t$
 9. $x = c_1e^t + (1/4)\cos t - (1/4)\sin t$;
 $y = -3c_1e^t - (3/4)\cos t - (1/4)\sin t$
 11. $u = c_1\cos 2t + c_2\sin 2t + c_3e^{\sqrt{3}t} + c_4e^{-\sqrt{3}t}$
 $- (3/10)e^t$;
 $v = c_1\cos 2t + c_2\sin 2t - (2/5)c_3e^{\sqrt{3}t}$
 $- (2/5)c_4e^{-\sqrt{3}t} + (1/5)e^t$

13. $x = 2c_2e^t\cos 2t - 2c_1e^t\sin 2t$;
 $y = c_1e^t\cos 2t + c_2e^t\sin 2t$
 15. $w = -(2/3)c_1e^{2t} + c_2e^{7t} + t + 1$;
 $z = c_1e^{2t} + c_2e^{7t} - 5t - 2$
 17. $x = c_1\cos t + c_2\sin t - 4c_3\cos(\sqrt{6}t)$
 $- 4c_4\sin(\sqrt{6}t)$;
 $y = c_1\cos t + c_2\sin t + c_3\cos(\sqrt{6}t) + c_4\sin(\sqrt{6}t)$
 19. $x = 2e^{3t} - e^{2t}$;
 $y = -2e^{3t} + 2e^{2t}$
 21. $x = e^t + e^{-t} + \cos t + \sin t$;
 $y = e^t + e^{-t} - \cos t - \sin t$
 23. Infinitely many solutions satisfying
 $x + y = e^t + e^{-2t}$
 25. $x(t) = -c_1e^t - 2c_2e^{2t} - c_3e^{3t}$;
 $y(t) = c_1e^t + c_2e^{2t} + c_3e^{3t}$;
 $z(t) = 2c_1e^t + 4c_2e^{2t} + 4c_3e^{3t}$
 27. $x(t) = c_1e^{8t} + c_2e^{4t} + c_3$;
 $y(t) = \frac{1}{2}(c_1e^{8t} - c_2e^{4t} + c_3)$;
 $z(t) = -c_1e^{8t} + c_3$
 29. $-3 \leq \lambda \leq -1$
 31. $x(t) = -\left(10 + \frac{20}{\sqrt{7}}\right)e^{r_1t} - \left(10 - \frac{20}{\sqrt{7}}\right)e^{r_2t}$
 $+ 20$ kg,
 $y(t) = \frac{30}{\sqrt{7}}e^{r_1t} - \frac{30}{\sqrt{7}}e^{r_2t} + 20$ kg, where
 $r_1 = \frac{-5 - \sqrt{7}}{100} \approx -0.0765$;
 $r_2 = \frac{-5 + \sqrt{7}}{100} \approx -0.0235$
 33. $x = \left[\frac{20 - 10\sqrt{5}}{\sqrt{5}}\right]e^{(-3+\sqrt{5})t/100}$
 $- \left[\frac{20 + 10\sqrt{5}}{\sqrt{5}}\right]e^{(-3-\sqrt{5})t/100} + 20$;
 $y = -\left(\frac{10}{\sqrt{5}}\right)e^{(-3+\sqrt{5})t/100} + \left(\frac{10}{\sqrt{5}}\right)e^{(-3-\sqrt{5})t/100} + 20$
 35. 90.4°F 37. $460/11 \approx 41.8^\circ\text{F}$

Exercises 5.3, page 259

1. $x'_1 = x_2$, $x'_2 = 3x_1 - tx_2 + t^2$;
 $x_1(0) = 3$, $x_2(0) = -6$
 3. $x'_1 = x_2$, $x'_2 = x_3$, $x'_3 = x_4$;
 $x'_4 = x_4 - 7x_1 + \cos t$; $x_1(0) = x_2(0) = 1$;
 $x_3(0) = 0$, $x_4(0) = 2$
 5. $x'_1 = x_2$, $x'_2 = x_2 - x_3 + 2t$;
 $x'_3 = x_4$, $x'_4 = x_1 - x_3 - 1$;
 $x_1(3) = 5$, $x_2(3) = 2$, $x_3(3) = 1$, $x_4(3) = -1$
 7. $x'_1 = x_2$, $x'_2 = x_3$, $x'_3 = x_4 + t$, $x'_4 = x_5$;
 $x'_5 = (2x_4 - 2x_3 + 1)/5$; $x_1(0) = x_2(0) = x_3(0) = 4$;
 $x_4(0) = x_5(0) = 1$

9. $t_{n+1} = t_n + h, \quad n = 0, 1, 2, \dots;$

$$x_{i,n+1} = x_{i,n} + \frac{h}{2} [f_i(t_n, x_{1,n}, \dots, x_{m,n}) + f_i(t_n + h, x_{1,n} + hf_1(t_n, x_{1,n}, \dots, x_{m,n}), \dots, x_{m,n} + hf_m(t_n, x_{1,n}, \dots, x_{m,n}))]$$

$i = 1, 2, \dots, m$

11. i	t_i	$y(t_i)$	13. i	t_i	$y(t_i)$
1	0.250	0.750000	1	0.250	0.25000
2	0.500	0.625000	2	0.500	0.50000
3	0.750	0.573529	3	0.750	0.75000
4	1.000	0.563603	4	1.000	1.00000

15. $y(1) \approx x_1(1; 2^{-2}) = 1.69, y'(1) \approx 1.82$

17. $u(1; 2^{-2}) = v(1; 2^{-2}) = 0.36789$

19. **Part (a)** **Part (b)** **Part (c)**

i	t_i	$x(t_i)$	$y(t_i)$	$x(t_i)$	$y(t_i)$	$x(t_i)$	$y(t_i)$
1	0.5	1.95247	2.25065	1.48118	2.42311	0.91390	2.79704
2	1.0	3.34588	1.83601	2.66294	1.45358	1.63657	1.13415
3	1.5	4.53662	3.36527	5.19629	2.40348	4.49334	1.07811
4	2.0	2.47788	4.32906	3.10706	4.64923	5.96115	5.47788
5	2.5	1.96093	2.71900	1.92574	3.32426	1.51830	5.93110
6	3.0	2.86412	1.96166	2.34143	2.05910	0.95601	2.18079
7	3.5	4.28449	2.77457	3.90106	2.18977	2.06006	0.98131
8	4.0	3.00965	4.11886	3.83241	3.89043	5.62642	1.38072
9	4.5	2.18643	3.14344	2.32171	3.79362	5.10594	5.10462
10	5.0	2.63187	2.25824	2.21926	2.49307	1.74187	5.02491

21. i	t_i	$x_1(t_i) \approx H(t_i)$
1	0.5	0.09573
2	1.0	0.37389
3	1.5	0.81045
4	2.0	1.37361
5	2.5	2.03111
6	3.0	2.75497
7	3.5	3.52322
8	4.0	4.31970
9	4.5	5.13307
10	5.0	5.95554

23. Yes, yes

25. $y(1) \approx x_1(1; 2^{-3}) = 1.25958$

27. $y(0.1) \approx 0.00647, \dots, y(2.0) \approx 1.60009$

29. (a) $P_1(10) \approx 0.567, P_2(10) \approx 0.463, P_3(10) \approx 0.463$

(b) $P_1(10) \approx 0.463, P_2(10) \approx 0.567, P_3(10) \approx 0.463$

(c) $P_1(10) \approx 0.463, P_2(10) \approx 0.463, P_3(10) \approx 0.567$

All populations approach 0.5.

Exercises 5.4, page 271

1. $x = y^3, \quad y > 0$



Figure B.21

3. $(1/\sqrt{2}, 1/\sqrt{2}), (-1/\sqrt{2}, -1/\sqrt{2})$ 5. $(0, 0)$

7. $e^x + ye^{-y} = c$

9. $e^x + xy - y^2 = c$

11. $y^2 - x^2 = c$

13. $(x-1)^2 + (y-1)^2 = c$

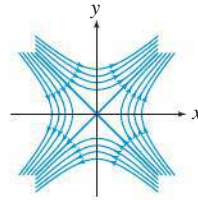


Figure B.22

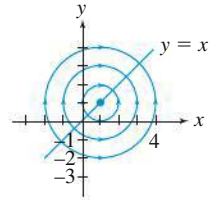


Figure B.23

15. $(-2, 1)$ is a saddle point (unstable).

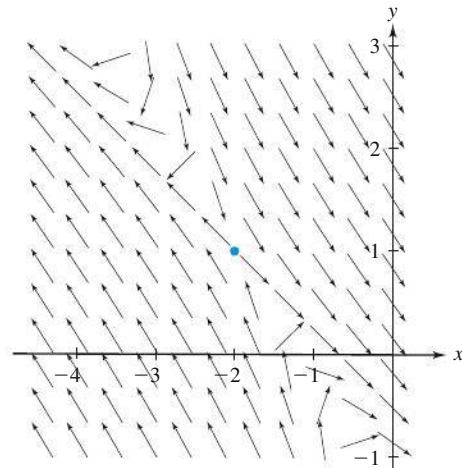


Figure B.24

17. $(0, 0)$ is a center (stable).

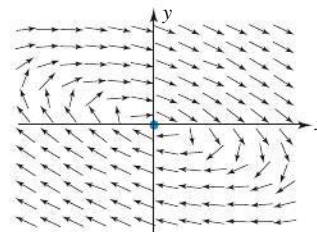


Figure B.25

19. $(0, 0)$ is a saddle point (unstable).

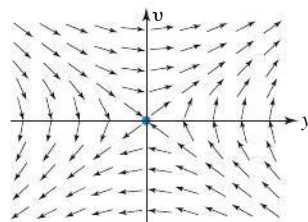


Figure B.26

21. $(0, 0)$ is a center (stable).

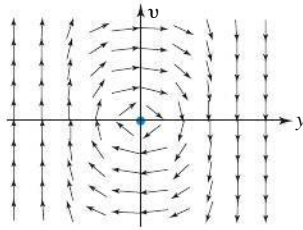


Figure B.27

23. $(0, 0)$ is a center (stable); $(1, 0)$ is a saddle point (unstable).

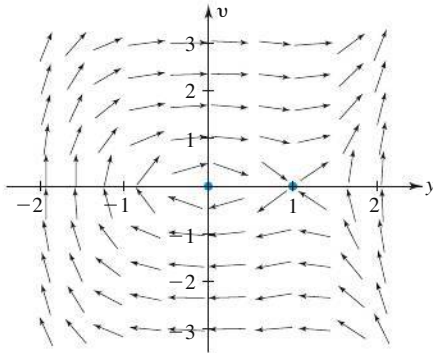


Figure B.28

25. (a) Periodic (b) Not periodic
(c) Critical point (periodic)

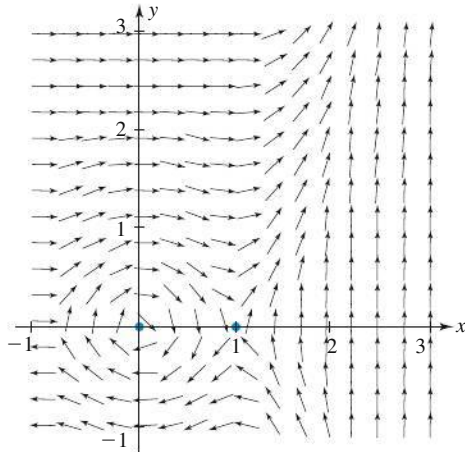


Figure B.29

27. $(x(t), y(t))$ approaches $(0, 0)$.

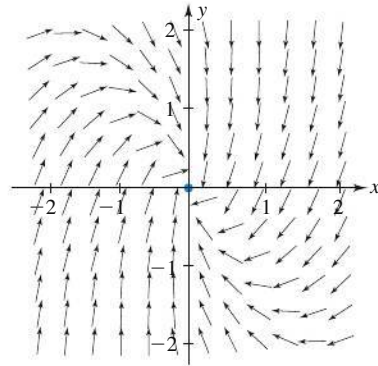


Figure B.30

29. Critical points are $(1, 0)$ and $(x, 1)$ with x arbitrary. Phase plane solutions are $x = 1$, and $y = C(x - 1)$. Solution starting at $(3, 2)$ goes to infinity. The others converge to $(1, 0)$.

31. (a) $y' = v$, $v' = f(y)$

(b) $\frac{dv}{dy} = \frac{f(y)}{v}$ implies $\int v dv = \int f(y) dy + \text{constant}$, or $\frac{v^2}{2} = F(y) + K$

33. (a) $x' = v$, $v' = -x + 1/(\lambda - x)$

(b) $\frac{dv}{dx} = \frac{-x + 1/(\lambda - x)}{v}$ implies

$$\int v dv = \int \left(-x + \frac{1}{\lambda - x} \right) dx + \text{constant}, \text{ or}$$

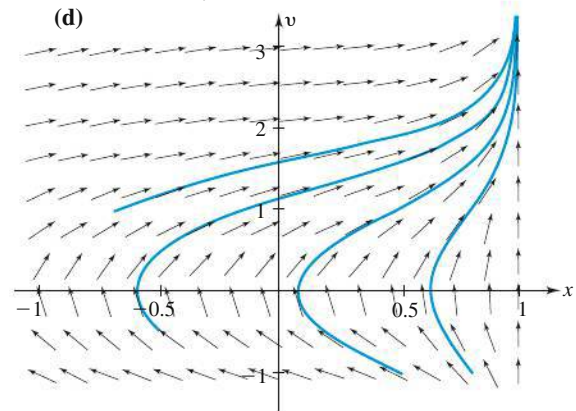
$$\frac{v^2}{2} = -\frac{x^2}{2} - \ln(\lambda - x) + \text{constant}. \text{ The solution equation follows with the choice } C/2 \text{ for the constant.}$$

- (c) At a critical point $v = 0$ and $-x + 1/(\lambda - x) = 0$.

$$\text{Solutions for the latter are } x = \frac{\lambda \pm \sqrt{\lambda^2 - 4}}{2}, \text{ and}$$

are real only for $\lambda \geq 2$.

- (d)

Figure B.31 $\lambda = 1$

- (e) See Figures B.31 and B.32. When $\lambda = 3$, one critical point is a center and the other is a saddle. For $\lambda = 1$, the bar is attracted to the magnet. For $\lambda = 3$, the bar may oscillate periodically, or (rarely) come to rest at the saddle critical point.

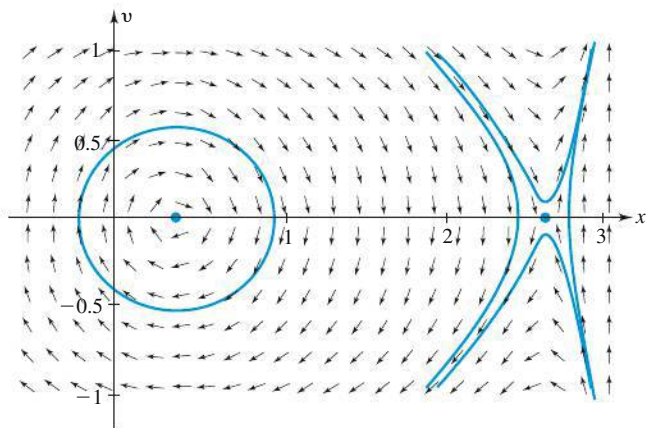


Figure B.32 $\lambda = 3$

35. (c) For upper half-plane, center is at $v = 0, y = -1$; for lower half-plane, center is at $v = 0, y = 1$.
 (d) All points on the segment $v = 0, -1 \leq y \leq 1$
 (e) $y = -0.5$; see Figure B.33.

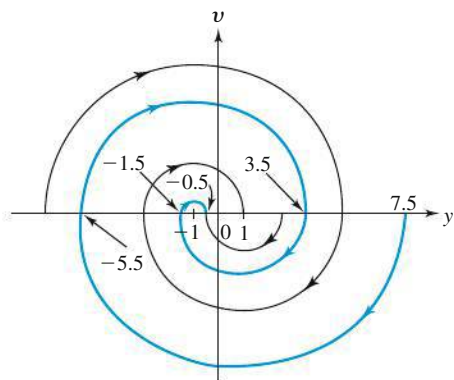


Figure B.33

Exercises 5.5, page 281

1. Runge–Kutta approximations

t_n	$p_n(r = 1.5)$	$p_n(r = 2)$	$p_n(r = 3)$
0.25	1.59600	1.54249	1.43911
0.5	2.37945	2.07410	1.64557
0.75	3.30824	2.47727	1.70801
1.0	4.30243	2.72769	1.72545
1.25	5.27054	2.86458	1.73024
1.5	6.13869	2.93427	1.73156
1.75	6.86600	2.96848	1.73192
2.0	7.44350	2.98498	1.73201
2.25	7.88372	2.99286	1.73204
2.5	8.20933	2.99661	1.73205
2.75	8.44497	2.99839	1.73205
3.0	8.61286	2.99924	1.73205
3.25	8.73117	2.99964	1.73205
3.5	8.81392	2.99983	1.73205
3.75	8.87147	2.99992	1.73205
4.0	8.91136	2.99996	1.73205
4.25	8.93893	2.99998	1.73205
4.5	8.95796	2.99999	1.73205
4.75	8.97107	3.00000	1.73205
5.0	8.98010	3.00000	1.73205

Limiting population is $3^{1/(r-1)}$.

$$3. \quad x(t) = \frac{1}{2} - \frac{2 \cos(\pi t/12)}{4 + (\pi/12)^2} - \frac{(\pi/12) \sin(\pi t/12)}{4 + (\pi/12)^2} + \left(\frac{19}{2} + \frac{2}{4 + (\pi/12)^2} \right) e^{-2t}$$

5. (a)

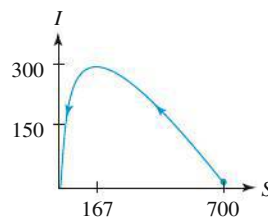


Figure B.34

(b) 295 persons

$$7. \quad P(t) = \exp\left\{ \frac{c[1 - \exp(-bt)]}{b} - bt \right\},$$

$$Q(t) = \exp\left\{ \frac{c[1 - \exp(-bt)]}{b} - bt \right\} [\exp(bt) - 1]$$

$$9. \quad N(t) = \frac{N_0(c+s)}{N_0s + (c+s-N_0s)e^{-(c+s)t}}, \quad N_0 := N(0)$$

11. Roughly, 2

Exercises 5.6, page 287

- $m_1 x'' = k_1(y - x)$,
 $m_2 y'' = -k_1(y - x) - k_2 y$;
 $x(0) = -1$, $x'(0) = 0$, $y(0) = 0$, $y'(0) = 0$
 $x(t) = -(8/17)\cos t - (9/17)\cos(\sqrt{20/3}t)$,
 $y(t) = -(6/17)\cos t + (6/17)\cos(\sqrt{20/3}t)$
- $mx'' = -kx + k(y - x)$,
 $my'' = -k(y - x) + k(z - y)$,
 $mz'' = -k(z - y) - kz$;
 The normal frequency $(1/2\pi)\sqrt{(2 + \sqrt{2})(k/m)}$ has the mode $x(t) = z(t) = -(1/\sqrt{2})y(t)$; the normal frequency $(1/2\pi)\sqrt{(2 - \sqrt{2})(k/m)}$ has the mode $x(t) = z(t) = (1/\sqrt{2})y(t)$; and the normal frequency $(1/2\pi)\sqrt{2k/m}$ has the mode $x(t) = -z(t)$, $y(t) \equiv 0$.
- $x(t) = -e^{-t} - te^{-t} - \cos t$; $y(t) = e^{-t} + te^{-t} - \cos t$
- $x(t) = (2/5)\cos t + (4/5)\sin t - (2/5)\cos\sqrt{6}t$
 $+ (\sqrt{6}/5)\sin\sqrt{6}t - \sin 2t$;
 $y(t) = (4/5)\cos t + (8/5)\sin t + (1/5)\cos\sqrt{6}t$
 $- (\sqrt{6}/10)\sin\sqrt{6}t - (1/2)\sin 2t$
- $(1/2\pi)\sqrt{g/l}$; $(1/2\pi)\sqrt{(g/l) + (2k/m)}$

Exercises 5.7, page 294

- $I(t) = (19/\sqrt{21})[e^{(-25-5\sqrt{21})t/2} - e^{(-25+5\sqrt{21})t/2}]$
- $I_p(t) = (4/51)\cos 20t - (1/51)\sin 20t$; resonance frequency is $5/\pi$.
- $M(\gamma) = 1/\sqrt{(100 - 4\gamma^2)^2 + 100\gamma^2}$

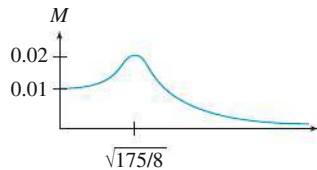


Figure B.35

- $L = 35 \text{ H}$, $R = 10 \text{ } \Omega$, $C = 1/15 \text{ F}$, and $E(t) = 50 \cos 10t \text{ V}$
- $I_1 = (3/5)e^{-3t/2} - (8/5)e^{-2t/3} + 1$,
 $I_2 = (1/5)e^{-3t/2} - (6/5)e^{-2t/3} + 1$,
 $I_3 = (2/5)e^{-3t/2} - (2/5)e^{-2t/3}$
- $(1/2)I_1' + 2q_3 = \cos 3t$ (where $I_3 = q_3'$),
 $(1/2)I_1' + I_2 = 0$, $I_1 = I_2 + I_3$;
 $I_1(0) = I_2(0) = I_3(0) = 0$;
 $I_1 = -(36/61)e^{-t}\cos\sqrt{3}t$
 $- (42\sqrt{3}/61)e^{-t}\sin\sqrt{3}t + (36/61)\cos 3t$
 $+ (30/61)\sin 3t$,
 $I_2 = (45/61)e^{-t}\cos\sqrt{3}t - (39\sqrt{3}/61)e^{-t}\sin\sqrt{3}t$
 $- (45/61)\cos 3t + (54/61)\sin 3t$,
 $I_3 = -(81/61)e^{-t}\cos\sqrt{3}t$
 $- (3\sqrt{3}/61)e^{-t}\sin\sqrt{3}t + (81/61)\cos 3t$
 $- (24/61)\sin 3t$

Exercises 5.8, page 303

- For $\omega = 3/2$: The Poincaré map alternates between the points $(0.8, 1.5)$ and $(0.8, -1.5)$. There is a subharmonic solution of period 4π . For $\omega = 3/5$: The Poincaré map cycles through the points $(-1.5625, 0.6)$, $(-2.1503, -0.4854)$, $(-0.6114, 0.1854)$, $(-2.5136, 0.1854)$, and $(-0.9747, -0.4854)$. There is a subharmonic solution of period 10π .
- The points become unbounded.
- The attractor is the point $(-1.0601, 0.2624)$.
- $\{1/7, 2/7, 4/7, 1/7, \dots\}$,
 $\{3/7, 6/7, 5/7, 3/7, \dots\}$
 - $\{1/15, 2/15, 4/15, 8/15, 1/15, \dots\}$,
 $\{1/5, 2/5, 4/5, 3/5, 1/5, \dots\}$,
 $\{1/3, 2/3, 1/3, \dots\}$,
 $\{7/15, 14/15, 13/15, 11/15, 7/15, \dots\}$
 - $x_n = 0$ for $n \geq j$

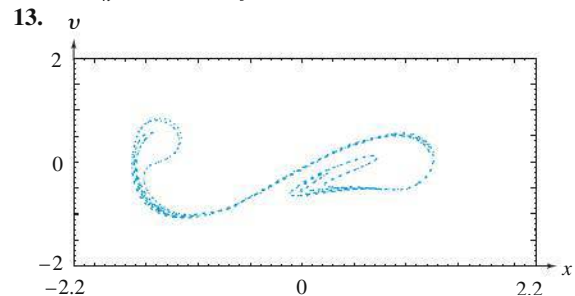


Figure B.36

Review Problems, page 306

- $x(t) = -(c_1/3)t^3 - (c_2/2)t^2 - (c_3 + 2c_1)t + c_4$,
 $y(t) = c_1 t^2 + c_2 t + c_3$
- $x(t) = c_1 \cos 3t + c_2 \sin 3t + e^t/10$,
 $y(t) = (3/2)(c_1 + c_2)\cos 3t - (3/2)(c_1 - c_2)\sin 3t$
 $- (11/20)e^t - (1/4)e^{-t}$
- $x(t) = 2 \sin t$, $y(t) = e^t - \cos t + \sin t$,
 $z(t) = e^t + \cos t + \sin t$
- $x(t) = -(13.9/4)e^{-t/6} - (4.9/4)e^{-t/2} + 4.8$,
 $y(t) = -(13.9/2)e^{-t/6} + (4.9/2)e^{-t/2} + 4.8$
- $x_1' = x_2$, $x_2' = x_3$, $x_3' = (1/3)(5 + e^t x_1 - 2x_2)$
- $x_1' = x_2$, $x_2' = x_3$, $x_3' = t - x_5 - x_6$,
 $x_4' = x_5$, $x_5' = x_6$, $x_6' = x_2 - x_3$
- $x^2 - (y - 1)^2 = c$; critical point $(0, 1)$ is saddle (unstable).
- Asymptotically stable spiral point
- $I_1 + I_2 + I_3 = 0$, $q/C = R_2 I_2$,
 $R_2 I_2 = R_1 I_3 + L dI_3/dt$, where q is the charge on the capacitor ($I_1 = dq/dt$);
 $I_3 = e^{-t}(A \cos t + B \sin t)$,
 $I_2 = e^{-t}(B \cos t - A \sin t)$,
 $I_1 = e^{-t}[(A - B)\sin t - (A + B)\cos t]$.

CHAPTER 6

Exercises 6.1, page 326

1. $(-\infty, 0)$ 3. $(3\pi/2, 5\pi/2)$ 5. $(0, \infty)$
7. Lin. indep. 9. Lin. dep.
11. Lin. indep. 13. Lin. indep.
15. $c_1e^{3x} + c_2e^{-x} + c_3e^{-4x}$
17. $c_1x + c_2x^2 + c_3x^3$
19. (a) $c_1e^x + c_2e^{-x}\cos 2x + c_3e^{-x}\sin 2x + x^2$
(b) $-e^x + e^{-x}\sin 2x + x^2$
21. (a) $c_1x + c_2x\ln x + c_3x(\ln x)^2 + \ln x$
(b) $3x - x\ln x + x(\ln x)^2 + \ln x$
23. (a) $2\sin x - x$
(b) $4x - 6\sin x$
29. (b) Let $f_1(x) = |x - 1|$ and $f_2(x) = x - 1$
33. e^{2x} , $(\sin x - 2\cos x)/5$, $-(2\sin x + \cos x)/5$
35. $xy''' - y'' + xy' - y = 0$

Exercises 6.2, page 332

1. $c_1 + c_2e^{2x} + c_3e^{-4x}$
3. $c_1e^{-x} + c_2e^{-2x/3} + c_3e^{x/2}$
5. $c_1e^{-x} + c_2e^{-x}\cos 5x + c_3e^{-x}\sin 5x$
7. $c_1e^{-x} + c_2e^{(3+\sqrt{65})x/4} + c_3e^{(3-\sqrt{65})x/4}$
9. $c_1e^{3x} + c_2xe^{3x} + c_3x^2e^{3x}$
11. $c_1e^{-x} + c_2xe^{-x} + c_3x^2e^{-x} + c_4x^3e^{-x}$
13. $c_1\cos\sqrt{2}x + c_2x\cos\sqrt{2}x + c_3\sin\sqrt{2}x + c_4x\sin\sqrt{2}x$
15. $c_1e^x + c_2xe^x + c_3e^{-3x} + (c_4 + c_5x)e^{-x}\cos 2x$
 $+ (c_6 + c_7x)e^{-x}\sin 2x$
17. $c_1e^{-4x} + c_2e^{3x} + (c_3 + c_4x + c_5x^2)e^{-2x}$
 $+ (c_6 + c_7x)e^{-2x}\cos x + (c_8 + c_9x)e^{-2x}\sin x + c_{10}$
 $+ c_{11}x + c_{12}x^2 + c_{13}x^3 + c_{14}x^4$
19. $e^x - 2e^{-2x} - 3e^{2x}$
21. $e^{2x} - \sqrt{2}e^x\sin\sqrt{2}x$
23. $x(t) = c_1 + c_2t + c_3e^t$, $y(t) = c_1 - c_2 + c_2t$
27. $c_1e^{1.120x} + c_2e^{0.296x} + c_3e^{-0.520x} + c_4e^{-2.896x}$
29. $c_1e^{-0.5x}\cos(0.866x) + c_2e^{-0.5x}\sin(0.866x)$
 $+ c_3e^{-0.5x}\cos(1.323x) + c_4e^{-0.5x}\sin(1.323x)$
31. (a) $\{x, x^{-1}, x^2\}$
(b) $\{x, x^2, x^{-1}, x^{-2}\}$
(c) $\{x, x^2\cos(3\ln x), x^2\sin(3\ln x)\}$
33. (b) $x(t) = c_1\cos t + c_2\sin t + c_3\cos\sqrt{6}t$
 $+ c_4\sin\sqrt{6}t$
(c) $y(t) = 2c_1\cos t + 2c_2\sin t - (c_3/2)\cos\sqrt{6}t$
 $- (c_4/2)\sin\sqrt{6}t$
(d) $x(t) = (3/5)\cos t + (2/5)\cos\sqrt{6}t$,
 $y(t) = (6/5)\cos t - (1/5)\cos\sqrt{6}t$
35. $c_1\cosh rx + c_2\sinh rx + c_3\cos rx + c_4\sin rx$, where
 $r^4 = k/(EI)$

Exercises 6.3, page 337

1. $c_1xe^x + c_2 + c_3x + c_4x^2$
3. $c_1x^2e^{-2x}$
5. $c_1e^x + c_2e^{3x} + c_3e^{-2x} - (1/6)xe^x + (1/6)x^2$
 $+ (5/18)x + 37/108$

7. $c_1e^x + c_2e^{-2x} + c_3xe^{-2x} - (1/6)x^2e^{-2x}$
9. $c_1e^x + c_2xe^x + c_3x^2e^x + (1/6)x^3e^x$
11. D^5
13. $D + 7$
15. $(D - 2)(D - 1)$
17. $[(D + 1)^2 + 4]^3$
19. $(D + 2)^2[(D + 5)^2 + 9]^2$
21. $c_3\cos 2x + c_4\sin 2x + c_5$
23. $c_3xe^{3x} + c_4x^2 + c_5x + c_6$
25. $c_3 + c_4x + c_5\cos 2x + c_6\sin 2x$
27. $c_3xe^{-x}\cos x + c_4xe^{-x}\sin x + c_5x^2 + c_6x + c_7$
29. $c_2x + c_3x^2 + c_6x^2e^x$
31. $-2e^{3x} + e^{-2x} + x^2 - 1$
33. $x^2e^{-2x} - x^2 + 3$
39. $x(t) = -(1/63)e^{3t} + c_1 + c_2t - c_3e^{\sqrt{2}t} - c_4e^{-\sqrt{2}t}$,
 $y(t) = (8/63)e^{3t} + c_1 + c_2t + c_3e^{\sqrt{2}t} + c_4e^{-\sqrt{2}t}$

Exercises 6.4, page 341

1. $(1/6)x^2e^{2x}$
3. $e^{2x}/16$
5. $\ln(\sec x) - (\sin x)\ln(\sec x + \tan x)$
7. $c_1x + c_2x^2 + c_3x^3 - (1/24)x^{-1}$
9. $-(1/2)e^x \int e^{-x}g(x)dx + (1/6)e^{-x} \int e^xg(x)dx$
 $+ (1/3)e^{2x} \int e^{-2x}g(x)dx$
11. $c_1x + c_2x^{-1} + c_3x^3 - x\sin x - 3\cos x + 3x^{-1}\sin x$

Review Problems, page 343

1. (a) $(0, \infty)$
(b) $(-4, -1)$, $(-1, 1)$, $(1, \infty)$
5. (a) $e^{-5x}(c_1 + c_2x) + e^{2x}(c_3 + c_4x + c_5x^2)$
 $+ (\cos x)(c_6 + c_7x) + (\sin x)(c_8 + c_9x)$
(b) $c_1 + c_2x + c_3x^2 + c_4x^3 + e^x(c_5 + c_6x)$
 $+ (e^{-x}\cos\sqrt{3}x)(c_7 + c_8x)$
 $+ (e^{-x}\sin\sqrt{3}x)(c_9 + c_{10}x)$
7. (a) D^3 (b) $D^2(D - 3)$ (c) $[D^2 + 4]^2$
(d) $[(D + 2)^2 + 9]^3$
(e) $D^3(D + 1)^2(D^2 + 4)(D^2 + 9)$
9. $c_1x + c_2x^5 + c_3x^{-1} - (1/21)x^{-2}$

CHAPTER 7

Exercises 7.2, page 360

1. $\frac{1}{s^2}$, $s > 0$ 3. $\frac{1}{s-6}$, $s > 6$
5. $\frac{s}{s^2 + 4}$, $s > 0$
7. $\frac{s-2}{(s-2)^2 + 9}$, $s > 2$
9. $e^{-2s}\left(\frac{2s+1}{s^2}\right)$, $s > 0$ 11. $\frac{e^{-\pi s} + 1}{s^2 + 1}$, all s

13. $\frac{6}{s+3} - \frac{2}{s^3} + \frac{2}{s^2} - \frac{8}{s}, \quad s > 0$
 15. $\frac{6}{s^4} - \frac{1}{(s-1)^2} + \frac{s-4}{(s-4)^2+1}, \quad s > 4$
 17. $\frac{6}{(s-3)^2+36} - \frac{6}{s^4} + \frac{1}{s-1}, \quad s > 3$
 19. $\frac{24}{(s-5)^5} - \frac{s-1}{(s-1)^2+7}, \quad s > 5$
 21. Continuous (hence piecewise continuous)
 23. Piecewise continuous
 25. Continuous (hence piecewise continuous)
 27. Neither
 29. All but functions (c), (e), and (h)

Exercises 7.3, page 365

1. $\frac{2}{s^3} + \frac{2}{(s-1)^2+4}$
 3. $\frac{s+1}{(s+1)^2+9} + \frac{1}{s-6} - \frac{1}{s}$
 5. $\frac{4}{(s+1)^3} - \frac{1}{s^2} + \frac{s}{s^2+16}$
 7. $\frac{24}{s^5} - \frac{24}{s^4} + \frac{12}{s^3} - \frac{4}{s^2} + \frac{1}{s}$
 9. $\frac{4(s+1)}{[(s+1)^2+4]^2}$ 11. $\frac{s}{s^2-b^2}$
 13. $\frac{1}{2s} - \frac{s}{2(s^2+4)}$ 15. $\frac{3s}{4(s^2+1)} + \frac{s}{4(s^2+9)}$
 17. $\frac{s}{2(s^2+9)} - \frac{s}{2(s^2+49)}$
 19. $\frac{n+m}{2[s^2+(n+m)^2]} + \frac{m-n}{2[s^2+(m-n)^2]}$
 21. $\frac{s-a}{(s-a)^2+b^2}$
 25. (a) $\frac{s^2-b^2}{(s^2+b^2)^2}$ (b) $\frac{2s^3-6sb^2}{(s^2+b^2)^3}$
 29. $\frac{1}{s^2+6s+10}$ 33. e^{-s}/s^2
 35. $e^{-(\pi/2)s}/(s^2+1)$

Exercises 7.4, page 374

1. $e^t t^3$ 3. $e^{-t} \cos 3t$ 5. $(1/2)e^{-2t} \sin 2t$
 7. $2e^{-2t} \cos 3t + 4e^{-2t} \sin 3t$
 9. $(3/2)e^t \cos 2t - 3e^t \sin 2t$
 11. $\frac{6}{s+5} - \frac{1}{s+2} - \frac{4}{s-1}$ 13. $\frac{1}{(s+1)^2} - \frac{2}{s}$
 15. $-\frac{3}{s+1} + \frac{(s-1)+2}{(s-1)^2+4}$
 17. $-\frac{5}{6s} + \frac{11}{10(s-2)} - \frac{4}{15(s+3)}$
 19. $\frac{1}{17} \left[\frac{1}{s-3} - \frac{s+1}{(s+1)^2+1} - \frac{4}{(s+1)^2+1} \right]$

21. $(1/3) + e^t + (14/3)e^{6t}$
 23. $-e^{-3t} + 2te^{-3t} + 6e^{-t}$
 25. $8e^{2t} - e^{-t} \cos 2t + 3e^{-t} \sin 2t$
 27. $-(5/3)e^{-t} + (5/12)e^{2t} + (5/4)e^{-2t}$
 29. $3e^{-2t} + 7e^t \cos t + 11e^t \sin t$
 31. $F_1(s) = F_2(s) = F_3(s) = 1/s^2$,
 $\mathcal{L}^{-1}\{1/s^2\}(t) = f_3(t) = t$
 33. $e^{5t}/t - e^{-2t}/t$
 35. $2(\cos t - \cos 3t)/t$
 39. $\frac{A}{s} + \frac{B}{s-1} + \frac{C}{s+2}$, where
 $A = \frac{2s+1}{(s-1)(s+2)} \Big|_{s=0} = \frac{-1}{2}$,
 $B = \frac{2s+1}{s(s+2)} \Big|_{s=1} = 1$,
 $C = \frac{2s+1}{s(s-1)} \Big|_{s=-2} = \frac{-1}{2}$
 41. $2e^{-t} - 4e^{3t} + 5e^{2t}$
 43. $\frac{4}{s+2} + \frac{2(s-1)+2(3)}{(s-1)^2+2^2}$

Exercises 7.5, page 382

1. $2e^t \cos 2t + e^t \sin 2t$ 3. $-e^{-3t} + 3te^{-3t}$
 5. $t^2 + \cos t - \sin t$ 7. $\cos t - 4e^{5t} + 8e^{2t}$
 9. $3(t-1)e^{t-1} - e^{-6(t-1)}$
 11. $2-t + e^{-t} + 2e^{-2}$
 13. $(7/5) \sin t + (11/5) \cos t + (3/5)e^{2t-\pi} - e^{-(t-\pi/2)}$
 15. $\frac{-s^2+s-1}{(s^2+1)(s-1)(s-2)}$
 17. $\frac{s^5+s^4+6}{s^4(s^2+s-1)}$ 19. $\frac{s^3+5s^2-6s+1}{s(s-1)(s^2+5s-1)}$
 21. $\frac{s^3+s^2+2s}{(s^2+1)(s-1)^2}$ 23. $\frac{-s^3+1+3se^{-2s}-e^{-2s}}{s^2(s^2+4)}$
 25. $2e^t - \cos t - \sin t$ 27. $(t^2-4)e^{-t}$
 29. $(3a-b)e^t/2 + (b-a)e^{3t}/2$
 31. $5/2 + (a-5/2)e^{-t} \cos t + (a+b-5/2)e^{-t} \sin t$
 35. $t^2/2$
 37. $\cos t + t \sin t + c(\sin t - t \cos t)$, (c arbitrary)
 39. $e(t) = -a \cos(\sqrt{k/I}t)$
 41. $e(t) = (-2aI/\sqrt{4Ik-\mu^2})e^{-\mu t/(2I)} \sin(\sqrt{4Ik-\mu^2}t/(2I))$

Exercises 7.6, page 390

1. $2e^{-s}/s^3$ 3. $e^{-2s}(4s^2+4s+2)/s^3$
 5. $(2e^{-s} - e^{-2s} + 2e^{-3s})/s$
 7. $[(e^{-s} - e^{-2s})(s+1)]/s^2$
 9. $(e^{-s} - 2e^{-2s} + e^{-3s})/s^2$
 11. $e^{t-2}u(t-2)$
 13. $e^{-2(t-2)}u(t-2) - 3e^{-2(t-4)}u(t-4)$
 15. $e^{-2(t-3)}[\cos(t-3) - 2 \sin(t-3)]u(t-3)$
 17. $(7e^{6-2t} - 6e^{3-t})u(t-3)$
 19. $10 - 10u(t-3\pi)[1 + e^{-(t-3\pi)}(\cos t + \sin t)]$
 $+ 10u(t-4\pi)[1 - e^{-(t-4\pi)}(\cos t + \sin t)]$

21. $\sin t + [1 - \cos(t-3)]u(t-3)$

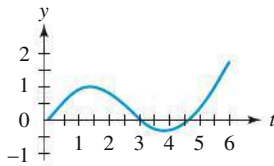


Figure B.37

23. $t + [4 - t + \sin(t-2) - 2\cos(t-2)]u(t-2)$

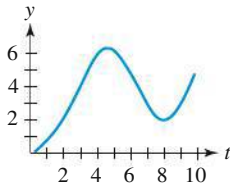


Figure B.38

25. $e^{-t}\cos t + 2e^{-t}\sin t$
 $+ (1/2)[1 - e^{2\pi-t}(\cos t + \sin t)]u(t-2\pi)$
 $- (1/2)[1 - e^{4\pi-t}(\cos t + \sin t)]u(t-4\pi)$
27. $e^{-t} + e^{-2t} + (1/2)[e^{-3t} - 2e^{-2(t+1)} + e^{-(t+4)}]u(t-2)$
29. $\cos 2t + (1/3)[1 - u(t-2\pi)]\sin t$
 $+ (1/6)[8 + u(t-2\pi)]\sin 2t$
31. $2e^{-2t} - 2e^{-3t}$
 $+ [1/36 + (1/6)(t-1) - (1/4)e^{-2(t-1)}]$
 $+ (2/9)e^{-3(t-1)}]u(t-1)$
 $- [19/36 + (1/6)(t-5)$
 $- (7/4)e^{-2(t-5)} + (11/9)e^{-3(t-5)}]u(t-5)$
33. $0.04 - 0.02 e^{-3t/125} + 0.02 u(t-10)[1 - e^{-3(t-10)/125}]$
35. The resulting differential equation has polynomial coefficients, so the Laplace transform method will result in a differential equation for the transform.

Exercises 7.7, page 396

1. $\frac{1 - 2se^{-2s} - e^{-2s}}{s^2(1 - e^{-2s})}$

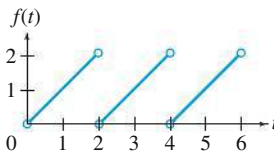


Figure B.39

3. $\frac{1}{1 - e^{-2s}} \left[\frac{1 - e^{-s-1}}{s+1} + \frac{e^{-s} - e^{-2s}}{s} \right]$



Figure B.40

5. $\frac{1}{s(1 + e^{-as})}$ 7. $\frac{1 - e^{-as}}{as^2(1 + e^{-as})}$

13. $\frac{e^{t-n}}{6} - \frac{e^{-t}}{2}(1 + e - e^{n+1})$
 $+ \frac{e^{-2t}}{3} \left[\frac{1 + e + e^2 - e^{2n+2}}{e+1} \right],$

for $n < t < n+1$

15. $\sum_{n=1}^{\infty} \frac{1}{s^n} = \frac{1}{s-1}$

17. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2ns^{2n}} = \frac{1}{2} \ln \left(1 + \frac{1}{s^2} \right)$

19. (a) $\sqrt{\pi/s}$

(b) $105\sqrt{\pi}/(16s^{9/2})$

27. (a) and (b), but not (c)

Exercises 7.8, page 404

1. $2te^t - e^t + \int_0^t e^{t-v}(t-v)g(v)dv$

3. $\int_0^t g(v)e^{2v-2t}\sin(t-v)dv + e^{-2t}\cos t + 3e^{-2t}\sin t$

5. $1 - \cos t$ 7. $2e^{5t} - 2e^{-2t}$ 9. $(t/2)\sin t$

11. $(2/3)e^{-2t} + (1/3)e^t$ 13. $s^{-2}(s-3)^{-1}$

15. $t/4 + (3/8)\sin 2t$ 17. $\cos t$ 19. 3

21. $e^{-t/2}\cos(\sqrt{3}t/2) - (1/\sqrt{3})e^{-t/2}\sin(\sqrt{3}t/2)$

23. $H(s) = (s^2 + 9)^{-1}$; $h(t) = (1/3)\sin 3t$;
 $y_k(t) = 2\cos 3t - \sin 3t$;

$y(t) = (1/3) \int_0^t [\sin 3(t-v)]g(v)dv + 2\cos 3t - \sin 3t$

25. $H(s) = (s^2 - s - 6)^{-1}$; $h(t) = (e^{3t} - e^{-2t})/5$;
 $y_k(t) = 2e^{3t} - e^{-2t}$;

$y(t) = (1/5) \int_0^t [e^{3(t-v)} - e^{-2(t-v)}]g(v)dv + 2e^{3t} - e^{-2t}$

27. $H(s) = (s^2 - 2s + 5)^{-1}$; $h(t) = (1/2)e^t\sin 2t$;
 $y_k(t) = e^t\sin 2t$;

$y(t) = (1/2) \int_0^t e^{(t-v)}[\sin 2(t-v)]g(v)dv + e^t\sin 2t$

29. $(1/30) \int_0^t e^{-2(t-v)}[\sin 6(t-v)]e(v)dv - e^{-2t}\cos 6t$
 $+ e^{-2t}\sin 6t$

31. $t^2/2$

Exercises 7.9, page 410

1. -1 3. -1 5. e^{-2} 7. $e^{-s} - e^{-3s}$

9. e^{-s} 11. 0 13. $-(\sin t)u(t-\pi)$

15. $e^t + e^{-3t} + (1/4)(e^{t-1} - e^{3-3t})u(t-1)$
 $- (1/4)(e^{t-2} - e^{6-3t})u(t-2)$

17. $2(e^{t-2} - e^{-(t-2)})u(t-2) + 2e^t - t^2 - 2$

19. $e^{-t} - e^{-5t} + (e/4)(e^{1-t} - e^{5-5t})u(t-1)$

21. $\sin t + (\sin t)u(t - 2\pi)$; see Fig. B.41.

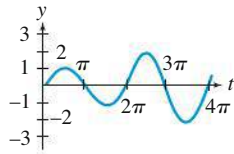


Figure B.41

23. $\sin t + (\sin t)u(t - \pi) + (\sin t)u(t - 2\pi)$

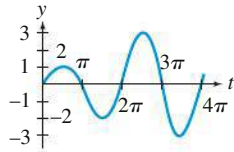


Figure B.42

25. $(1/2)e^{-2t}\sin 2t$ 27. $(1/2)e^t\sin 2t$
 29. The mass remains stopped at $x(t) \equiv 0$, $t > \pi/2$
 35. $\frac{L}{6EI}[3\lambda x^2 - x^3 + (x - \lambda)^3 u(x - \lambda)]$

Exercises 7.10, page 413

1. $x = e^t$; $y = e^t$ 3. $z = e^t$; $w \equiv 0$
 5. $x = (7/4)e^t + (7/4)e^{-t} - (3/2)\cos t$;
 $y = (7/4)e^t - (7/4)e^{-t} + (1/2)\sin t$
 7. $x = -(150/17)e^{5t/2}\cos(\sqrt{15}t/2)$
 $- (334\sqrt{15}/85)e^{5t/2}\sin(\sqrt{15}t/2) - (3/17)e^{-3t}$;
 $y = (46/17)e^{5t/2}\cos(\sqrt{15}t/2)$
 $- (146\sqrt{15}/85)e^{5t/2}\sin(\sqrt{15}t/2) + (22/17)e^{-3t}$
 9. $x = 4e^{-2t} - e^{-t} - \cos t$; $y = 5e^{-2t} - e^{-t}$
 11. $x = (e^t - e^{-t})/2 - (1/2)[e^{t-2} - e^{-(t-2)}]u(t-2)$;
 $y = 1 - (e^t + e^{-t})/2$
 $- [1 - (e^{t-2} + e^{-(t-2)})/2]u(t-2)$
 13. $x = e^{-t} - (1/2)[e^{-(t-\pi)} + \cos t - \sin t]u(t-\pi)$;
 $y = e^{-t} + [1 - (1/2)e^{-(t-\pi)} + (1/2)\cos t$
 $+ (1/2)\sin t]u(t-\pi)$
 15. $x = t^2$; $y = t - 1$
 17. $x = (t-2)e^{t-2}$; $y = e^{t-2}$
 19. $x = -7e^{-t} + e^t$; $y = 2e^{-t}$; $z = -13e^{-t} + e^t$
 21. $x = 0.01(-e^{-t/2} + e^{-t/6}) + [0.48 - 0.36e^{-(t-5)/6}$
 $- 0.12e^{-(t-5)/2}]u(t-5)$;
 $y = 0.02(e^{-t/2} + e^{-t/6}) + [0.48 - 0.72e^{-(t-5)/6}$
 $+ 0.24e^{-(t-5)/2}]u(t-5)$
 23. $2I_1 + (0.1)I_3' + (0.2)I_1' = 6$, $(0.1)I_3' - I_2 = 0$,
 $I_1 = I_2 + I_3$; $I_1(0) = I_2(0) = I_3(0) = 0$;
 $I_1 = -e^{-20t} - 2e^{-5t} + 3$,
 $I_2 = -2e^{-20t} + 2e^{-5t}$,
 $I_3 = e^{-20t} - 4e^{-5t} + 3$

Review Problems, page 415

1. $\frac{3}{s} + e^{-2s}\left[\frac{1}{s} - \frac{1}{s^2}\right]$ 3. $\frac{2}{(s+9)^3}$
 5. $\frac{1}{s-2} - \frac{6}{s^4} + \frac{2}{s^3} - \frac{5}{s^2+25}$ 7. $\frac{s^2-36}{(s^2+36)^2}$
 9. $2e^{-4s}\left[\frac{1}{s^3} + \frac{4}{s^2} + \frac{8}{s}\right]$ 11. $(7/2)t^2e^{-3t}$
 13. $2e^t + 2e^{-2t}\cos 3t + e^{-2t}\sin 3t$
 15. $e^{-2t} + e^{-t} - 2te^{-t}$
 17. $[2e^{t-2} + 2e^{4-2t}]u(t-2)$
 19. $e^{2t} - e^{5t}$
 21. $-(3/2) + t + t^2/2 + (3/2)e^{-t}\cos t - (1/2)e^{-t}\sin t$
 23. $(2/\sqrt{7})e^{-3t/2}\sin(\sqrt{7}t/2)$
 $+ \{(1/4) - [3/(4\sqrt{7})]\}e^{-3(t-1)/2}\sin[\sqrt{7}(t-1)/2]$
 $- (1/4)e^{-3(t-1)/2}\cos[\sqrt{7}(t-1)/2]\}u(t-1)$
 25. $c[t + te^{-2t} + e^{-2t} - 1]$
 27. $(9/10)e^{-3t} + (1/10)\cos t - (3/10)\sin t$
 29. $(s^2 - 5s + 6)^{-1}$; $e^{3t} - e^{2t}$
 31. $x = 1 - (e^t + e^{-t})/2$
 $- [1 - (e^{t-2} + e^{-(t-2)})/2]u(t-2)$,
 $y = (e^t - e^{-t})/2 - (1/2)[e^{t-2} - e^{-(t-2)}]u(t-2)$

CHAPTER 8

Exercises 8.1, page 425

1. $1 + x + x^2 + \dots$
 3. $x + x^2 + (1/2)x^3 + \dots$
 5. $1 - (1/6)t^3 + (1/180)t^6 + \dots$
 7. $(1/6)\theta^3 - (1/120)\theta^5 + (1/5040)\theta^7 + \dots$
 9. (a) $p_3(x) = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3$
 (b) $|\varepsilon_3(1.5)| = |f^{(4)}(\xi)(1.5-1)^4/4!| \leq 6(0.5)^4/4!$
 (c) $|\ln(1.5) - p_3(1.5)| = 0.011202\dots$
 (d)

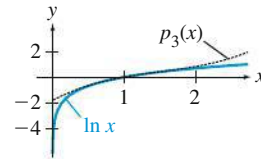


Figure B.43

13. $t + (1/2)t^2 - (1/6)t^3 + \dots$ 15. $1 - x^2/6$

Exercises 8.2, page 433

1. $[-1, 3]$ 3. $(-4, 0)$ 5. $[1, 3]$
 9. $\sum_{n=0}^{\infty} \left[\frac{1}{n+1} + 2^{-n-1} \right] x^n$
 11. $x + x^2 + (1/3)x^3 + \dots$

13. $1 - 2x + (5/2)x^2 + \dots$

15. (c) $1 - (1/2)x + (1/4)x^2 - (1/24)x^3 + \dots$

17. $\sum_{n=1}^{\infty} (-1)^n n x^{n-1}$ 19. $\sum_{k=1}^{\infty} a_k 2k x^{2k-1}$

21. $\ln(x+1) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1}$

23. $\sum_{k=0}^{\infty} (k+1) a_{k+1} x^k$

25. $\sum_{k=1}^{\infty} a_{k-1} x^k$ 29. $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} (x-\pi)^{2n}}{(2n)!}$

31. $1 + \sum_{n=1}^{\infty} 2x^n$

33. $6(x-1) + 3(x-1)^2 + (x-1)^3$

35. (a) $\sum_{n=0}^{\infty} (-1)^n (x-1)^n$ (b) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} (x-1)^n$

Exercises 8.3, page 443

1. -1 3. $\pm\sqrt{2}$ 5. $-1, 2$

7. $x = n\pi$, n an integer

9. $\theta \leq 0$ and $\theta = n\pi$, $n = 1, 2, 3, \dots$

11. $y = a_0(1 - 2x + (3/2)x^2 - x^3/3 + \dots)$

13. $a_0(1 + x^4/12 + \dots) + a_1(x + x^5/20 + \dots)$

15. $a_0(1 - x^2/2 - x^3/6 + \dots)$
 $+ a_1(x + x^2/2 - x^3/6 + \dots)$

17. $a_0(1 - x^2/2 + \dots) + a_1(x - x^3/6 + \dots)$

19. $a_0 \sum_{n=0}^{\infty} \frac{1}{n!} x^{2n} = a_0 e^{x^2}$

21. $a_0(1 - 2x^2 + x^4/3)$
 $+ a_1 \left[x + \sum_{k=1}^{\infty} \frac{(-3)(-1) \cdots (2k-5)}{(2k+1)!} x^{2k+1} \right]$

29. (a) $a_0 \left[1 + \sum_{k=1}^{\infty} (-1)^k \frac{n(n-2)(n-4) \cdots (n-2k+2)(n+1)(n+3) \cdots (n+2k-1)}{(2k)!} x^{2k} \right]$
 $+ a_1 \left[x + \sum_{k=1}^{\infty} (-1)^k \frac{(n-1)(n-3) \cdots (n-2k+1)(n+2)(n+4) \cdots (n+2k)}{(2k+1)!} x^{2k+1} \right]$

(c) $P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = (1/2)(3x^2 - 1)$

31. (a) $1 - (1/2)t^2 + (\eta/6)t^3 + [(1 - \eta^2)/24]t^4 + \dots$ (b) Yes

23. $a_{3k+2} = 0$, $k = 0, 1, \dots$

$$a_0 \left(1 + \sum_{k=1}^{\infty} \frac{[1 \cdot 4 \cdot 7 \cdots (3k-2)]^2}{(3k)!} x^{3k} \right) + a_1 \left(x + \sum_{k=1}^{\infty} \frac{[2 \cdot 5 \cdot 8 \cdots (3k-1)]^2}{(3k+1)!} x^{3k+1} \right)$$

25. $2 + x^2 - (5/12)x^4 + (11/72)x^6 + \dots$

27. $x + (1/6)x^3 - (1/12)x^4 + (7/120)x^5 + \dots$

29. $1 - 2x + x^2 - (1/6)x^3$

31. $1 + 2x - (3/4)x^2 - (5/6)x^3$

35. $10 - 25t^2 + (250/3)t^3 - (775/4)t^4 + \dots$

Exercises 8.4, page 449

1. 2 3. $\sqrt{3}$ 5. $\pi/2$

7. $a_0[1 - (x-1)^2 + (1/2)(x-1)^4 - (1/6)(x-1)^6 + \dots]$

9. $a_0[1 + (x-1)^2 + \dots]$
 $+ a_1[(x-1) + (1/3)(x-1)^3 + \dots]$

11. $a_0[1 - (1/8)(x-2)^2 + (1/32)(x-2)^3 + \dots]$
 $+ a_1[(x-2) + (1/8)(x-2)^2 - (7/96)(x-2)^3 + \dots]$

13. $1 - (1/2)t^2 + (1/6)t^4 - (31/720)t^6 + \dots$

15. $1 + x + (1/24)x^4 + (1/60)x^5 + \dots$

17. $1 - (1/6)(x-\pi)^3 + (1/120)(x-\pi)^5 - (1/180)(x-\pi)^6 + \dots$

19. $-1 + x + x^2 + (1/2)x^3 + \dots$

21. $a_0[1 + (1/2)x^2 + (1/8)x^4 + (1/48)x^6 + \dots]$
 $+ [(1/2)x^2 + (1/12)x^4 + (11/720)x^6 + \dots]$

23. $a_0[1 - (1/2)x^2 + \dots] + a_1[x - (1/3)x^3 + \dots]$
 $+ [(1/2)x^2 + (1/3)x^3 + \dots]$

25. $a_0[1 - (1/2)x^2 + \dots] + a_1[x + \dots]$
 $+ [(1/2)x^2 - (1/6)x^3 + \dots]$

27. $a_0[1 - (1/2)x^2 - (1/6)x^3 + \dots]$
 $+ a_1[x + (1/2)x^2 + \dots] + [(1/6)x^3 + \dots]$

Exercises 8.5, page 453

1. $c_1x^{-2} + c_2x^{-3}$
3. $c_1x \cos(4 \ln x) + c_2x \sin(4 \ln x)$
5. $c_1x^3 \cos(2 \ln x) + c_2x^3 \sin(2 \ln x)$
7. $c_1x + c_2x^{-1} \cos(3 \ln x) + c_3x^{-1} \sin(3 \ln x)$
9. $c_1x + c_2x^{-1/2} \cos\left[\left(\sqrt{19}/2\right) \ln x\right] + c_3x^{-1/2} \sin\left[\left(\sqrt{19}/2\right) \ln x\right]$
11. $c_1(x-3)^{-2} + c_2(x-3)^{1/2}$
13. $c_1x + c_2x^2 + (4/15)x^{-1/2}$
15. $2t^4 + t^{-3}$
17. $(31/17)x + (3/17)x^{-2} \cos(5 \ln x) - (76/85)x^{-2} \sin(5 \ln x)$

Exercises 8.6, page 464

1. ± 1 are regular.
3. $\pm i$ are regular.
5. 1 is regular, -1 is irregular.
7. 2 is regular, -1 is irregular.
9. -4 is regular and 2 is irregular.
11. $r^2 - 3r - 10 = 0$; $r_1 = 5$, $r_2 = -2$
13. $r^2 - 5r/9 - 4/3 = 0$; $r_1 = (5 + \sqrt{457})/18$, $r_2 = (5 - \sqrt{457})/18$
15. $r^2 - 1 = 0$; $r_1 = 1$, $r_2 = -1$
17. $r^2 + r - 12 = 0$; $r_1 = 3$, $r_2 = -4$
19. $a_0[x^{2/3} - (1/2)x^{5/3} + (5/28)x^{8/3} - (1/21)x^{11/3} + \dots]$
21. $a_0[1 - (1/4)x^2 + (1/64)x^4 - (1/2304)x^6 + \dots]$
23. $a_0[x - (1/3)x^2 + (1/12)x^3 - (1/60)x^4 + \dots]$
25. $a_0 \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+(3/2)}}{2^{n-1}(n+2)!}$
27. $a_0 \sum_{n=0}^{\infty} \frac{x^{2n+2}}{2^{2n}(n+1)n!}$
29. a_0x^2
31. $a_0 \sum_{n=0}^{\infty} \frac{x^n}{n!} = a_0e^x$; yes, $a_0 < 0$
33. $a_0 \left[x^{1/3} + \sum_{n=1}^{\infty} \frac{2^{n-2}(3n+4)x^{n+(1/3)}}{3^n n!} \right]$; yes, $a_0 < 0$
35. $a_0[x^{5/6} - (1/11)x^{11/6} + (1/374)x^{17/6} - (1/25,806)x^{23/6} + \dots]$
37. $a_0[x^{4/3} + (1/17)x^{7/3} + (1/782)x^{10/3} + (1/68,034)x^{13/3} + \dots]$
39. The expansion $\sum_{n=0}^{\infty} n!x^n$ diverges for $x \neq 0$.
41. The transformed equation is $z d^2y/dz^2 + 3 dy/dz - y = 0$. Also $zp(z) = 3$ and $z^2q(z) = -z$ are analytic at $z = 0$; hence, $z = 0$ is a regular singular point.
 $y_1(x) = a_0[1 + (1/3)x^{-1} + (1/24)x^{-2} + (1/360)x^{-3} + \dots]$
45. $a_0[x^{-3} - (1/2)x^{-1} - (1/8)x - (1/144)x^3 + \dots] + a_3[1 + (1/10)x^2 + (1/280)x^4 + (1/15,120)x^6 + \dots]$

Exercises 8.7, page 473

1. $c_1y_1(x) + c_2y_2(x)$, where $y_1(x) = x^{2/3} - (1/2)x^{5/3} + (5/28)x^{8/3} + \dots$ and $y_2(x) = x^{1/3} - (1/2)x^{4/3} + (1/5)x^{7/3} + \dots$
3. $c_1y_1(x) + c_2y_2(x)$, where $y_1(x) = 1 - (1/4)x^2 + (1/64)x^4 + \dots$ and $y_2(x) = y_1(x) \ln x + (1/4)x^2 - (3/128)x^4 + (11/13,824)x^6 + \dots$
5. $c_1[x - (1/3)x^2 + (1/12)x^3 + \dots] + c_2[x^{-1} - 1]$
7. $c_1y_1(x) + c_2y_2(x)$, where $y_1(x) = x^{3/2} - (1/6)x^{5/2} + (1/48)x^{7/2} + \dots$ and $y_2(x) = x^{-1/2} - (1/2)x^{1/2}$
9. $c_1w_1 + c_2w_2$, where $w_1(x) = x^2 + (1/8)x^4 + (1/192)x^6 + \dots$ and $w_2(x) = w_1(x) \ln x + 2 - (3/32)x^4 - (7/1152)x^6 + \dots$
11. $c_1y_1(x) + c_2y_2(x)$, where $y_1(x) = x^2$ and $y_2(x) = x^2 \ln x - 1 + 2x - (1/3)x^3 + (1/24)x^4 + \dots$
13. $c_1y_1(x) + c_2y_2(x)$, where $y_1(x) = 1 + x + (1/2)x^2 + \dots$ and $y_2(x) = y_1(x) \ln x - [x + (3/4)x^2 + (11/36)x^3 + \dots]$
15. $c_1y_1(x) + c_2y_2(x)$, where $y_1(x) = x^{1/3} + (7/6)x^{4/3} + (5/9)x^{7/3} + \dots$ and $y_2(x) = 1 + 2x + (6/5)x^2 + \dots$; all solutions are bounded near the origin.
17. $c_1y_1(x) + c_2y_2(x) + c_3y_3(x)$, where $y_1(x) = x^{5/6} - (1/11)x^{11/6} + (1/374)x^{17/6} + \dots$, $y_2(x) = 1 - x + (1/14)x^2 + \dots$, and $y_3(x) = y_2(x) \ln x + 7x - (117/196)x^2 + (4997/298116)x^3 + \dots$
19. $c_1y_1(x) + c_2y_2(x) + c_3y_3(x)$, where $y_1(x) = x^{4/3} + (1/17)x^{7/3} + (1/782)x^{10/3} + \dots$, $y_2(x) = 1 - (1/3)x - (1/30)x^2 + \dots$, and $y_3(x) = x^{-1/2} - (1/5)x^{1/2} - (1/10)x^{3/2} + \dots$
21. (b) $c_1y_1(t) + c_2y_2(t)$, where $y_1(t) = 1 - (1/6)(\alpha t)^2 + (1/120)(\alpha t)^4 + \dots$, and $y_2(t) = t^{-1}[1 - (1/2)(\alpha t)^2 + (1/24)(\alpha t)^4 + \dots]$
 (c) $c_1y_1(x) + c_2y_2(x)$, where $y_1(x) = 1 - (1/6)(\alpha/x)^2 + (1/120)(\alpha/x)^4 + \dots$, and $y_2(x) = x[1 - (1/2)(\alpha/x)^2 + (1/24)(\alpha/x)^4 + \dots]$
23. $c_1y_1(x) + c_2y_2(x)$, where $y_1(x) = 1 + 2x + 2x^2$ and $y_2(x) = -(1/6)x^{-1} - (1/24)x^{-2} - (1/120)x^{-3} + \dots$
25. For $n = 0$, $c_1 + c_2[\ln x + x + (1/4)x^2 + (1/18)x^3 + \dots]$; and for $n = 1$, $c_1(1-x) + c_2[(1-x) \ln x + 3x - (1/4)x^2 - (1/36)x^3 + \dots]$

Exercises 8.8, page 485

1. $c_1F\left(1, 2; \frac{1}{2}; x\right) + c_2x^{1/2}F\left(\frac{3}{2}, \frac{5}{2}; \frac{3}{2}; x\right)$
3. $c_1F\left(1, 1; \frac{1}{2}; x\right) + c_2x^{1/2}F\left(\frac{3}{2}, \frac{3}{2}; \frac{3}{2}; x\right)$

5. $F(1, 1; 2; x) = \sum_{n=0}^{\infty} \frac{1}{(n+1)} x^n = -x^{-1} \ln(1-x)$
7. $F\left(\frac{1}{2}, 1; \frac{3}{2}; x^2\right) = \sum_{n=0}^{\infty} (1/2)_n x^{2n} / (3/2)_n$
 $= \sum_{n=0}^{\infty} x^{2n} / (2n+1)$
 $= \frac{1}{2} x^{-1} \ln\left(\frac{1+x}{1-x}\right)$
9. $(1-x)^{-1}, (1-x)^{-1} \ln x$
13. $c_1 J_{1/2}(x) + c_2 J_{-1/2}(x)$ 15. $c_1 J_1(x) + c_2 Y_1(x)$
17. $c_1 J_{2/3}(x) + c_2 J_{-2/3}(x)$
19. $J_1(x) \ln x - x^{-1} + (3/64)x^3 - (7/2304)x^5 + \dots$
21. $x^{3/2} J_{3/2}(x)$
27. $J_0(x) \ln x + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2^{2n}(n!)^2} \left[1 + \frac{1}{2} + \dots + \frac{1}{n}\right] x^{2n}$
29. $1, x, (3x^2 - 1)/2, (5x^3 - 3x)/2, (35x^4 - 30x^2 + 3)/8$
37. $1, 2x, 4x^2 - 2, 8x^3 - 12x$
39. $1, 1-x, (2-4x+x^2)/2, (6-18x+9x^2-x^3)/6$
41. (b) $z = c_1 \cos x + c_2 \sin x$, so for x large,
 $y(x) \approx c_1 x^{-1/2} \cos x + c_2 x^{-1/2} \sin x$

Review Problems, page 489

1. (a) $1-x + (3/2)x^2 - (5/3)x^3 + \dots$
 (b) $-1+x - (1/6)x^3 + (1/6)x^4 + \dots$
3. (a) $a_0[1+x^2+\dots] + a_1[x + (1/3)x^3 + \dots]$
 (b) $a_0[1 + (1/2)x^2 - (1/6)x^3 + \dots] + a_1[x - (1/2)x^2 + (1/2)x^3 + \dots]$
5. $a_0[1 + (1/2)(x-2)^2 - (1/24)(x-2)^4 + \dots] + a_1(x-2)$
7. (a) $a_0[x^3 + x^4 + (1/4)x^5 + (1/36)x^6 + \dots]$
 (b) $a_0[x - (1/4)x^2 + (1/20)x^3 - (1/120)x^4 + \dots]$
9. (a) $c_1 y_1(x) + c_2 y_2(x)$, where
 $y_1(x) = x + x^2 + (1/2)x^3 + \dots = xe^x$ and
 $y_2(x) = y_1(x) \ln x - x^2 - (3/4)x^3 - (11/36)x^4 + \dots$
 (b) $c_1 y_1(x) + c_2 y_2(x)$, where
 $y_1(x) = 1 + 2x + x^2 + \dots$ and
 $y_2(x) = y_1(x) \ln x - [4x + 3x^2 + (22/27)x^3 + \dots]$
 (c) $c_1 y_1(x) + c_2 y_2(x)$, where
 $y_1(x) = 1 - (1/6)x + (1/96)x^2 + \dots$ and
 $y_2(x) = y_1(x) \ln x - 8x^{-2} - 4x^{-1} + (29/36) + \dots$
 (d) $c_1 y_1(x) + c_2 y_2(x)$, where
 $y_1(x) = x^2 - (1/4)x^3 + (1/40)x^4 + \dots$ and
 $y_2(x) = x^{-1} + (1/2) + (1/4)x + \dots$

CHAPTER 9

Exercises 9.1, page 500

1. $\begin{bmatrix} x \\ y \end{bmatrix}' = \begin{bmatrix} 7 & 2 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$
3. $\begin{bmatrix} x \\ y \\ z \end{bmatrix}' = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ 0 & 4 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

5. $\begin{bmatrix} x \\ y \end{bmatrix}' = \begin{bmatrix} \sin t & e^t \\ \cos t & a + bt^3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$
7. $\begin{bmatrix} y \\ v \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ -k/m & -b/m \end{bmatrix} \begin{bmatrix} y \\ v \end{bmatrix}$
9. $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ t & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}; x_1 = y, x_2 = y'$
11. $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -3 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix};$
 $x_1 = x, x_2 = x', x_3 = y, x_4 = y'$
13. $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}' = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ \cos t & 3 & -t^2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -e^t & t & 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix};$
 $x_1 = x, x_2 = x', x_3 = y, x_4 = y', x_5 = y''$

Exercises 9.2, page 504

1. $x_1 = 2, x_2 = 1, x_3 = 1$
3. $x_1 = -s + t, x_2 = s, x_3 = t$
5. $x_1 = 0, x_2 = 0$
7. $x_1 = 3s, x_2 = s \quad (-\infty < s < \infty)$
9. $x_1 = 2s, x_2 = (-1+i)s$, with s any complex number
11. $x_1 = -(s+1)/2, x_2 = -(11s+1)/2, x_3 = s \quad (-\infty < s < \infty)$
13. For $r = 2$, unique solution is $x_1 = x_2 = 0$.
 For $r = 1$, solutions are $x_1 = 3s, x_2 = s, -\infty < s < \infty$.

Exercises 9.3, page 513

1. (a) $\begin{bmatrix} 1 & 1 \\ 5 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 7 & 3 \\ 7 & 18 \end{bmatrix}$
3. (a) $\begin{bmatrix} 18 & 14 \\ 4 & 5 \end{bmatrix}$ (b) $\begin{bmatrix} 8 & 12 \\ 3 & 5 \end{bmatrix}$ (c) $\begin{bmatrix} 16 & 3 \\ 5 & 19 \end{bmatrix}$
5. (a) $\begin{bmatrix} -1 & -2 \\ -1 & -3 \end{bmatrix}$ (b) $\begin{bmatrix} -5 & -1 \\ -8 & -1 \end{bmatrix}$ (c) $\begin{bmatrix} -6 & -3 \\ -9 & -4 \end{bmatrix}$
7. (c) Yes (d) Yes 9. $\begin{bmatrix} 4/9 & -1/9 \\ 1/9 & 2/9 \end{bmatrix}$ 11. Doesn't exist
13. $\begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & -2 \\ 1 & 1 & 0 \end{bmatrix}$ 17. $\begin{bmatrix} (4/3)e^{-t} & -(1/3)e^{-t} \\ -(1/3)e^{-4t} & (1/3)e^{-4t} \end{bmatrix}$
19. $\begin{bmatrix} e^{-t} & (1/2)e^{-t} & -(1/2)e^{-t} \\ (1/3)e^t & -(1/2)e^t & (1/6)e^t \\ -(1/3)e^{-2t} & 0 & (1/3)e^{-2t} \end{bmatrix}$
21. 11 23. -12 25. 25 27. 2, 3
29. 0, 1, 1 31. $\begin{bmatrix} 3e^{3t} \\ 6e^{3t} \\ -3e^{3t} \end{bmatrix}$ 33. $\begin{bmatrix} 5e^{5t} & 6e^{2t} \\ -10e^{5t} & -2e^{2t} \end{bmatrix}$

39. (a) $\begin{bmatrix} (1/2)t^2 + c_1 & e^t + c_2 \\ t + c_3 & e^t + c_4 \end{bmatrix}$
 (b) $\begin{bmatrix} \sin 1 & -1 + \cos 1 \\ 1 - \cos 1 & \sin 1 \end{bmatrix}$
 (c) $\begin{bmatrix} (1 + e^t) \cos t + (e^t - t) \sin t & (e^t - t) \cos t - (e^t + 1) \sin t \\ (e^t - 1) \sin t + e^t \cos t & (e^t - 1) \cos t - e^t \sin t \end{bmatrix}$

Exercises 9.4, page 521

1. $\begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} + \begin{bmatrix} t^2 \\ e^t \end{bmatrix}$
 3. $\begin{bmatrix} dx/dt \\ dy/dt \\ dz/dt \end{bmatrix} = \begin{bmatrix} t^2 & -1 & -1 \\ 0 & 0 & e^t \\ t & -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} t \\ 5 \\ -e^t \end{bmatrix}$
 5. $\begin{bmatrix} x_1'(t) \\ x_2'(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 10 & 3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \sin t \end{bmatrix};$
 $x_1 = y, \quad x_2 = y'$
 7. $\begin{bmatrix} x_1'(t) \\ x_2'(t) \\ x_3'(t) \\ x_4'(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ t^2 \end{bmatrix};$
 $x_1 = w, \quad x_2 = w', \quad x_3 = w'', \quad x_4 = w'''$
 9. $x_1'(t) = 5x_1(t) + 2e^{-2t};$
 $x_2'(t) = -2x_1(t) + 4x_2(t) - 3e^{-2t}$
 11. $x_1'(t) = x_1(t) + x_3(t) + e^t;$
 $x_2'(t) = -x_1(t) + 2x_2(t) + 5x_3(t) + t;$
 $x_3'(t) = 5x_2(t) + x_3(t)$
 13. LI 15. LD 17. LI 19. LI
 21. Not a fundamental solution set
 23. Yes; $\begin{bmatrix} e^{-t} & e^t & e^{3t} \\ 2e^{-t} & 0 & -e^{3t} \\ e^{-t} & e^t & 2e^{3t} \end{bmatrix};$
 $c_1 \begin{bmatrix} e^{-t} \\ 2e^{-t} \\ e^{-t} \end{bmatrix} + c_2 \begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix} + c_3 \begin{bmatrix} e^{3t} \\ -e^{3t} \\ 2e^{3t} \end{bmatrix}$
 25. $\frac{3}{2} \begin{bmatrix} te^t \\ te^t \end{bmatrix} - \frac{1}{4} \begin{bmatrix} e^t \\ 3e^t \end{bmatrix} + \begin{bmatrix} t \\ 2t \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c_1 \begin{bmatrix} e^t \\ e^t \end{bmatrix} + c_2 \begin{bmatrix} e^{-t} \\ 3e^{-t} \end{bmatrix}$
 29. $\mathbf{X}^{-1}(t) = \begin{bmatrix} 0 & (1/4)e^t & -(1/4)e^t \\ -(1/5)e^{2t} & (4/5)e^{2t} & -(2/5)e^{2t} \\ (1/5)e^{-3t} & (9/20)e^{-3t} & (3/20)e^{-3t} \end{bmatrix};$
 $\mathbf{x}(t) = \begin{bmatrix} -(3/2)e^{-t} + (3/5)e^{-2t} - (1/10)e^{3t} \\ (1/4)e^{-t} - (1/5)e^{-2t} - (1/20)e^{3t} \\ (5/4)e^{-t} - (1/5)e^{-2t} - (1/20)e^{3t} \end{bmatrix}$
 37. (b) $2e^{6t}$ (e) $2e^{6t}$; same

Exercises 9.5, page 531

1. Eigenvalues are $r_1 = 0$ and $r_2 = -5$ with associated eigenvectors $\mathbf{u}_1 = \begin{bmatrix} s \\ 2s \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 2s \\ -s \end{bmatrix}$.
 3. Eigenvalues are $r_1 = 2$ and $r_2 = 3$ with associated eigenvectors $\mathbf{u}_1 = s \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{u}_2 = s \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

5. Eigenvalues are $r_1 = 1, r_2 = 2$, and $r_3 = -2$ with

$$\text{associated eigenvectors } \mathbf{u}_1 = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{u}_2 = s \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix},$$

$$\text{and } \mathbf{u}_3 = s \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}.$$

7. Eigenvalues are $r_1 = 1, r_2 = 2$, and $r_3 = 5$ with

$$\text{associated eigenvectors } \mathbf{u}_1 = s \begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix}, \mathbf{u}_2 = s \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix},$$

$$\text{and } \mathbf{u}_3 = s \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

9. Eigenvalues are $r_1 = i$ and $r_2 = -i$, with associated

$$\text{eigenvectors } \mathbf{u}_1 = s \begin{bmatrix} i \\ 1 \end{bmatrix}, \text{ and } \mathbf{u}_2 = s \begin{bmatrix} -i \\ 1 \end{bmatrix},$$

$$11. c_1 e^{3t/2} \begin{bmatrix} 3 \\ 10 \end{bmatrix} + c_2 e^{t/2} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$13. c_1 e^{-t} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} + c_3 e^{-5t} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$15. c_1 e^{-t} \begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix} + c_2 e^t \begin{bmatrix} 1 \\ -6 \\ 4 \end{bmatrix} + c_3 e^{4t} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

17. (b)

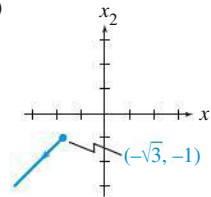


Figure B.43

- (c)

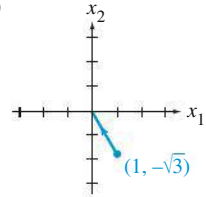


Figure B.44

- (d)

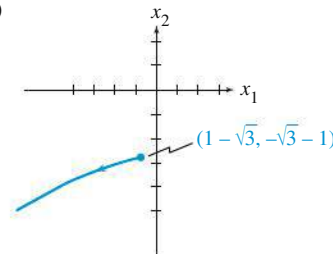


Figure B.45

$$19. \begin{bmatrix} e^{3t} & e^{-3t} \\ 4e^{3t} & -2e^{-3t} \end{bmatrix} \quad 21. \begin{bmatrix} e^t & e^{2t} & e^{4t} \\ e^t & 2e^{2t} & 4e^{4t} \\ e^t & 4e^{2t} & 16e^{4t} \end{bmatrix}$$

$$23. \begin{bmatrix} e^{2t} & e^{-t} & e^{3t} & -e^{7t} \\ 0 & -3e^{-t} & 0 & e^{7t} \\ 0 & 0 & e^{3t} & 2e^{7t} \\ 0 & 0 & 0 & 8e^{7t} \end{bmatrix}$$

$$\begin{aligned} 25. \quad x &= -c_1 e^t - 2c_2 e^{2t} - c_3 e^{3t}, \\ y &= c_1 e^t + c_2 e^{2t} + c_3 e^{3t}, \\ z &= 2c_1 e^t + 4c_2 e^{2t} + 4c_3 e^{3t} \end{aligned}$$

$$27. \begin{bmatrix} e^{-0.3473t} & e^{0.5237t} & 0.0286e^{-7.0764t} \\ -0.3157e^{-0.3473t} & 0.4761e^{0.5237t} & -0.1837e^{-7.0764t} \\ 0.0844e^{-0.3473t} & 0.1918e^{0.5237t} & e^{-7.0764t} \end{bmatrix}$$

$$29. \begin{bmatrix} 0.0251e^{3.4142t} & -0.2361e^{-1.6180t} & e^{0.6180t} & e^{0.5858t} \\ 0.0858e^{3.4142t} & 0.3820e^{-1.6180t} & 0.6180e^{0.6180t} & 0.5858e^{0.5858t} \\ 0.2929e^{3.4142t} & -0.6180e^{-1.6180t} & 0.3820e^{0.6180t} & 0.3431e^{0.5858t} \\ e^{3.4142t} & e^{-1.6180t} & 0.2361e^{0.6180t} & 0.2010e^{0.5858t} \end{bmatrix}$$

$$31. \begin{bmatrix} 2e^{4t} + e^{-2t} \\ 2e^{4t} - e^{-2t} \end{bmatrix} \quad 33. \begin{bmatrix} -3e^{-t} + e^{5t} \\ -2e^{-t} - e^{5t} \\ e^{-t} + e^{5t} \end{bmatrix}$$

$$35. \text{(b)} \mathbf{x}_1(t) = e^{-t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{(c)} \mathbf{x}_2(t) = te^{-t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$37. \text{(b)} \mathbf{x}_1(t) = e^{2t} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{(c)} \mathbf{x}_2(t) = te^{2t} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + e^{2t} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{(d)} \mathbf{x}_3(t) = \frac{t^2}{2} e^{2t} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + te^{2t} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + e^{2t} \begin{bmatrix} 0 \\ -6/5 \\ 1/5 \end{bmatrix}$$

$$39. \text{(b)} \mathbf{x}_1(t) = e^t \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}; \quad \mathbf{x}_2(t) = e^t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{(c)} \mathbf{x}_3(t) = te^t \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} + e^t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{(d)} \mathbf{0}$$

$$43. c_1 \begin{bmatrix} 3t^2 \\ t^2 \end{bmatrix} + c_2 \begin{bmatrix} t^4 \\ t^4 \end{bmatrix}$$

$$45. x_1(t) = (2.5/2)(e^{-3t/25} + e^{-t/25}), \\ x_2(t) = (2.5)(e^{-t/25} - e^{-3t/25})$$

$$47. \text{(a)} r_1 = 2.39091, \quad r_2 = -2.94338, \quad r_3 = 3.55247$$

$$\text{(b)} \mathbf{u}_1 = \begin{bmatrix} 1 \\ -2.64178 \\ -9.31625 \end{bmatrix}; \quad \mathbf{u}_2 = \begin{bmatrix} 1 \\ -1.16825 \\ 0.43862 \end{bmatrix};$$

$$\mathbf{u}_3 = \begin{bmatrix} 1 \\ 0.81004 \\ -0.12236 \end{bmatrix}$$

$$\text{(c)} c_1 e^{r_1 t} \mathbf{u}_1 + c_2 e^{r_2 t} \mathbf{u}_2 + c_3 e^{r_3 t} \mathbf{u}_3, \text{ where the } r_i\text{'s and the } \mathbf{u}_i\text{'s are given in parts (a) and (b).}$$

Exercises 9.6, page 537

$$1. c_1 \begin{bmatrix} 2 \cos 2t \\ \cos 2t + \sin 2t \end{bmatrix} + c_2 \begin{bmatrix} 2 \sin 2t \\ \sin 2t - \cos 2t \end{bmatrix}$$

$$3. c_1 e^t \cos t \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - c_1 e^t \sin t \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + c_2 e^t \sin t \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c_2 e^t \cos t \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + c_3 e^t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Note: Equivalent answer is obtained if $\text{col}(-1, 1, 0)$ and $\text{col}(-2, 0, 1)$ are replaced by $\text{col}(-5, 1, 2)$ and $\text{col}(0, -2, 1)$, respectively.

$$5. \begin{bmatrix} e^{-t} \cos 4t & e^{-t} \sin 4t \\ 2e^{-t} \sin 4t & -2e^{-t} \cos 4t \end{bmatrix} \quad 7. \begin{bmatrix} 1 & \cos t & \sin t \\ 0 & -\cos t & -\sin t \\ 0 & -\sin t & \cos t \end{bmatrix}$$

$$9. \begin{bmatrix} \cos t & \sin t & 0 & 0 \\ -\sin t & \cos t & 0 & 0 \\ 0 & 0 & \cos t & \sin t \\ 0 & 0 & -\sin t & \cos t \end{bmatrix} \quad 11. \begin{bmatrix} e^t \sin t - e^t \cos t & -e^t \sin t - e^t \cos t & -\cos t & -\sin t \\ 2e^t \sin t & -2e^t \cos t & \sin t & -\cos t \\ 2e^t \sin t + 2e^t \cos t & 2e^t \sin t - 2e^t \cos t & \cos t & \sin t \\ 4e^t \cos t & 4e^t \sin t & -\sin t & \cos t \end{bmatrix}$$

$$13. \text{(a)} \begin{bmatrix} e^{-2t}(\sin t - \cos t) \\ -2e^{-2t} \sin t \end{bmatrix} \quad \text{(b)} \begin{bmatrix} -e^{-2(t-\pi)} \cos t \\ e^{-2(t-\pi)}(\cos t - \sin t) \end{bmatrix} \\ \text{(c)} \begin{bmatrix} e^{-2(t+2\pi)}(2 \cos t - 3 \sin t) \\ e^{-2(t+2\pi)}(\cos t + 5 \sin t) \end{bmatrix} \quad \text{(d)} \begin{bmatrix} e^{\pi-2t} \cos t \\ e^{\pi-2t}(\sin t - \cos t) \end{bmatrix}$$

$$17. c_1 \begin{bmatrix} t^{-1} \\ 0 \\ -2t^{-1} \end{bmatrix} + c_2 \begin{bmatrix} t^{-1} \cos(\ln t) \\ t^{-1} \sin(\ln t) \\ -t^{-1} \cos(\ln t) \end{bmatrix} + c_3 \begin{bmatrix} t^{-1} \sin(\ln t) \\ -t^{-1} \cos(\ln t) \\ -t^{-1} \sin(\ln t) \end{bmatrix}$$

$$19. \frac{\sqrt{9-\sqrt{17}}}{2\sqrt{2}\pi} \approx 0.249; \quad \frac{\sqrt{9+\sqrt{17}}}{2\sqrt{2}\pi} \approx 0.408$$

$$21. I_1 = (5/6)e^{-2t} \sin 3t, \\ I_2 = -(10/13)e^{-2t} \cos 3t + (25/78)e^{-2t} \sin 3t, \\ I_3 = (10/13)e^{-2t} \cos 3t + (20/39)e^{-2t} \sin 3t$$

Exercises 9.7, page 542

1. $c_1 e^{7t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} 1 \\ -4 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \end{bmatrix}$
3. $c_1 e^{-3t} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + c_2 e^{3t} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_3 e^{3t} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + e^t \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$
5. $t \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} + \frac{e^{2t}}{3} \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + \frac{\sin t}{2} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \frac{\cos t}{2} \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$
 $+ c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 e^t \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c_3 e^t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$
7. $\mathbf{x}_p = t\mathbf{a} + \mathbf{b} + \sin 3t \mathbf{c} + \cos 3t \mathbf{d}$
9. $\mathbf{x}(t) = t\mathbf{a} + \mathbf{b} + e^{2t}\mathbf{c} + (\sin t)\mathbf{d} + (\cos t)\mathbf{e}$
11. $c_1 \begin{bmatrix} \sin t \\ \cos t \end{bmatrix} + c_2 \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix}$
13. $c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 e^{6t} \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \begin{bmatrix} t^{-1}/2 \\ t^{-1} \end{bmatrix}$
15. $c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 e^{-5t} \begin{bmatrix} -2 \\ 1 \end{bmatrix} + \begin{bmatrix} \ln|t| + (8/5)t - 8/25 \\ 2 \ln|t| + (16/5)t + 4/25 \end{bmatrix}$
17. $c_1 e^{2t} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + c_3 e^{-t} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + e^t \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$
19. $\begin{bmatrix} c_1 \cos t + c_2 \sin t + 1 \\ -c_1 \sin t + c_2 \cos t - t \\ c_3 e^t + c_4 e^{-t} - (1/4)e^{-t} - t + (1/2)te^{-t} \\ c_3 e^t - c_4 e^{-t} - (1/4)e^{-t} - 1 - (1/2)te^{-t} \end{bmatrix}$

Exercises 9.8, page 551

1. (a) $r = 3$; $k = 2$ (b) $e^{3t} \begin{bmatrix} 1 & -2t \\ 0 & 1 \end{bmatrix}$
3. (a) $r = -1$; $k = 3$ (b) $e^{-t} \begin{bmatrix} 1 + 3t - (3/2)t^2 & t & -t + (1/2)t^2 \\ -3t & 1 & t \\ 9t - (9/2)t^2 & 3t & 1 - 3t + (3/2)t^2 \end{bmatrix}$
5. (a) $r = -2$; $k = 2$ (b) $e^{-2t} \begin{bmatrix} 1 & 0 & 0 \\ 4t & 1 & 0 \\ t & 0 & 1 \end{bmatrix}$
7. $\begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$
9. $\frac{1}{2} \begin{bmatrix} e^t + \cos t - \sin t & 2 \sin t & e^t - \cos t - \sin t \\ e^t - \sin t - \cos t & 2 \cos t & e^t + \sin t - \cos t \\ e^t - \cos t + \sin t & -2 \sin t & e^t + \cos t + \sin t \end{bmatrix}$
11. $\frac{1}{25} \begin{bmatrix} -4 + 29e^{5t} - 20te^{5t} & 20 - 20e^{5t} & -8 + 8e^{5t} - 40te^{5t} \\ -5 + 5e^{5t} & 25 & -10 + 10e^{5t} \\ 2 - 2e^{5t} + 10te^{5t} & -10 + 10e^{5t} & 4 + 21e^{5t} + 20te^{5t} \end{bmatrix}$
13. $\begin{bmatrix} (1 - t + t^2/2)e^t & (t - t^2)e^t & (t^2/2)e^t & 0 & 0 \\ (t^2/2)e^t & (1 - t + t^2)e^t & (t + t^2/2)e^t & 0 & 0 \\ (t + t^2/2)e^t & (-3t - t^2)e^t & (1 + 2t + t^2/2)e^t & 0 & 0 \\ 0 & 0 & 0 & \cos t & \sin t \\ 0 & 0 & 0 & -\sin t & \cos t \end{bmatrix}$

21. (a) $\begin{bmatrix} 4te^t + 5e^t \\ 2te^t + 4e^t \end{bmatrix}$
- (b) $\begin{bmatrix} -2e^{t-1} + 2e^{2(t-1)} - 3e^{2t-1} + (4t-1)e^t \\ -e^{t-1} + 2e^{2(t-1)} - 3e^{2t-1} + (2t+1)e^t \end{bmatrix}$
- (c) $\begin{bmatrix} (-20 + 2e^{-5})e^t + (-3e^{-5} - e^{-10})e^{2t} + (4t+3)e^t \\ (-10 + e^{-5})e^t + (-3e^{-5} - e^{-10})e^{2t} + (2t+3)e^t \end{bmatrix}$
- (d) $\begin{bmatrix} -18e^{t+1} + 14e^{2t+2} - 3e^{2t+1} + (4t+7)e^t \\ -9e^{t+1} + 14e^{2t+2} - 3e^{2t+1} + (2t+5)e^t \end{bmatrix}$
23. $x = (c_1 + c_2t)e^t + c_3e^{-t} - t^2 - 4t - 6$;
 $y = -c_2e^t - 2c_3e^{-t} - t^2 - 2t - 3$
25. (a) $\left\{ \begin{bmatrix} e^t \\ e^t \end{bmatrix}, \begin{bmatrix} e^{2t} \\ 2e^{2t} \end{bmatrix} \right\}$ (c) $te^t \begin{bmatrix} 2 \\ 2 \end{bmatrix} + e^t \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- (d) $c_1 e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + te^t \begin{bmatrix} 2 \\ 2 \end{bmatrix} + e^t \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
27. $c_1 e^{2t} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + c_3 e^{-t} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$
 $+ te^{-t} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + e^{-t} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$
29. $c_1 t \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 t^{-1} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$
 $+ \begin{bmatrix} -(3/4)t^{-1} - (1/2)t^{-1} \ln t + 1 \\ -(3/4)t^{-1} - (3/2)t^{-1} \ln t + 2 \end{bmatrix}$
33. $I_1 = I_2 + 3I_5$, $I_3 = 3I_5$, $I_4 = 2I_5$, where
 $I_2 = (1/(20\sqrt{817})) \left[(13 - \sqrt{817})e^{-(31+\sqrt{817})5t/2} \right.$
 $\left. - (13 + \sqrt{817})e^{(-31+\sqrt{817})5t/2} \right] + 1/10$,
 $I_5 = (3/(40\sqrt{817})) \left[(31 - \sqrt{817})e^{-(31+\sqrt{817})5t/2} \right.$
 $\left. - (31 + \sqrt{817})e^{(-31+\sqrt{817})5t/2} \right] + 3/20$

$$15. \begin{bmatrix} (1+t+t^2/2)e^{-t} & (t+t^2)e^{-t} & (t^2/2)e^{-t} & 0 & 0 \\ (-t^2/2)e^{-t} & (1+t-t^2)e^{-t} & (t-t^2/2)e^{-t} & 0 & 0 \\ (-t+t^2/2)e^{-t} & (-3t+t^2)e^{-t} & (1-2t+t^2/2)e^{-t} & 0 & 0 \\ 0 & 0 & 0 & (1+2t)e^{-2t} & te^{-2t} \\ 0 & 0 & 0 & -4te^{-2t} & (1-2t)e^{-2t} \end{bmatrix}$$

$$17. c_1 e^{-t} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} -t \\ -1+t \\ 2-t \end{bmatrix} + c_3 e^{-2t} \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \quad 19. c_1 e^{2t} \begin{bmatrix} 3 \\ 6 \\ 1 \\ 1 \end{bmatrix} + c_2 e^t \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_3 e^t \begin{bmatrix} 1 \\ t \\ 0 \\ 0 \end{bmatrix} + c_4 e^t \begin{bmatrix} 2t \\ t+t^2 \\ 0 \\ 1 \end{bmatrix}$$

$$21. \begin{bmatrix} e^{-2t} \\ 4te^{-2t} + e^{-2t} \\ te^{-2t} - e^{-2t} \end{bmatrix}, \quad 23. e^{-t} \begin{bmatrix} 3t \\ 3 \\ 9t \end{bmatrix} + e^{At} \begin{bmatrix} 2 - e^t(t^2 - 2t + 2) \\ 1 + e^t(t - 1) \\ 6 - 3e^t(t^2 - 2t + 2) \end{bmatrix}, \text{ where } e^{At} \text{ is the matrix in the answer to Problem 3.}$$

$$29. \mathbf{x}(t) = \begin{bmatrix} (4/13)e^{-3t} + 6te^{3t} + (2/13)(3\sin 2t - 2\cos 2t) \\ e^{3t} \\ (8/13)e^{-3t} + (3t-1)e^{3t} + (4/13)(3\sin 2t - 2\cos 2t) \end{bmatrix}.$$

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$$1. c_1 e^{3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{4t} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad 3. c_1 e^{-t} \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} + c_3 e^{3t} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_4 e^{3t} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \quad 5. \begin{bmatrix} e^{2t} & e^{3t} \\ -e^{2t} & -2e^{3t} \end{bmatrix}$$

$$7. c_1 e^{-t} \begin{bmatrix} 1 \\ -2 \end{bmatrix} + c_2 e^{3t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1/3 \\ -14/3 \end{bmatrix} \quad 9. c_1 e^{-t} \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} t \\ 1 \\ 3t \end{bmatrix} + c_3 e^{-t} \begin{bmatrix} -t + (1/2)t^2 \\ t \\ 1 - 3t + (3/2)t^2 \end{bmatrix} + \begin{bmatrix} 2+t \\ 7-3t \\ 10 \end{bmatrix}$$

$$11. \begin{bmatrix} 3e^t - 2e^{2t} \\ 3e^t - 4e^{2t} \end{bmatrix} \quad 13. c_1 t^{-1} \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} + c_2 t^{-1} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + c_3 t^4 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$15. \begin{bmatrix} (1/2)e^t + (2/3)e^{5t} - (1/6)e^{-t} & -e^t + (2/3)e^{5t} + (1/3)e^{-t} & -(1/2)e^t + (2/3)e^{5t} - (1/6)e^{-t} \\ (1/3)e^{5t} - (1/3)e^{-t} & (1/3)e^{5t} + (2/3)e^{-t} & (1/3)e^{5t} - (1/3)e^{-t} \\ -(1/2)e^t + (1/2)e^{-t} & e^t - e^{-t} & (1/2)e^t + (1/2)e^{-t} \end{bmatrix}$$

17. (a) false (b) false (c) false (d) true (e) true (f) false (g) true

CHAPTER 10

Exercises 10.2, page 570

1. $y = [4/(e - e^{-1})](e^{-x} - e^x)$ 3. $y \equiv 0$
5. $y = e^x + 2x - 1$ 7. $y = \cos x + c \sin x$; c arbitrary
9. $\lambda_n = (2n-1)^2/4$ and $y_n = c_n \sin[(2n-1)x/2]$, where $n = 1, 2, 3, \dots$ and the c_n 's are arbitrary.
11. $\lambda_n = n^2$, $n = 0, 1, 2, \dots$; $y_0 = a_0$ and $y_n = a_n \cos nx + b_n \sin nx$, $n = 1, 2, 3, \dots$, where a_0 , a_n , and b_n are arbitrary.
13. The eigenvalues are the roots of $\tan(\sqrt{\lambda_n} \pi) + \sqrt{\lambda_n} = 0$, where $\lambda_n > 0$. For n large, $\lambda_n \approx (2n-1)^2/4$, n is a positive integer. The eigenfunctions are $y_n = c_n [\sin(\sqrt{\lambda_n} x) + \sqrt{\lambda_n} \cos(\sqrt{\lambda_n} x)]$, where the c_n 's are arbitrary.
15. $u(x, t) = e^{-3t} \sin x - 6e^{-48t} \sin 4x$
17. $u(x, t) = e^{-3t} \sin x - 7e^{-27t} \sin 3x + e^{-75t} \sin 5x$
19. $u(x, t) = 3 \cos 6t \sin 2x + 12 \cos 39t \sin 13x$
21. $u(x, t) = 6 \cos 6t \sin 2x + 2 \cos 18t \sin 6x + (11/27) \sin 27t \sin 9x - (14/45) \sin 45t \sin 15x$
23. $u(x, t) = \sum_{n=1}^{\infty} n^{-2} e^{-2\pi^2 n^2 t} \sin n\pi x$
25. If $K > 0$, then $T(t)$ becomes unbounded as $t \rightarrow \infty$, and so the temperature $u(x, t) = X(x)T(t)$ becomes unbounded at each position x . Since the temperature must remain bounded for all time, $K \not> 0$.
33. (a) $u(x) \equiv 50$ (b) $u(x) = 30x/L + 10$

Exercises 10.3, page 584

1. Odd 3. Even 5. Neither

$$9. f(x) \sim \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n} \sin nx$$

$$11. f(x) \sim 1 + \sum_{n=1}^{\infty} \left[\frac{2}{\pi^2 n^2} (-1 + (-1)^n) \cos \frac{n\pi x}{2} + \frac{1}{\pi n} ((-1)^{n+1} - 1) \sin \frac{n\pi x}{2} \right]$$

$$13. f(x) \sim \frac{1}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2 \pi^2} \cos n\pi x$$

$$15. f(x) \sim [(\sinh \pi)/\pi] \left(1 + 2 \sum_{n=1}^{\infty} \left[\frac{(-1)^n}{1+n^2} \cos nx + \frac{(-1)^{n+1}n}{1+n^2} \sin nx \right] \right)$$

$$17. \text{ The } 2\pi\text{-periodic function } g(x), \text{ where}$$

$$g(x) = \begin{cases} x & (-\pi < x < \pi), \\ 0 & (x = \pm \pi) \end{cases}$$

$$19. \text{ The } 4\text{-periodic function } g(x), \text{ where}$$

$$g(x) = \begin{cases} 1 & (-2 < x < 0), \\ x & (0 < x < 2), \\ 1/2 & (x = 0), \\ 3/2 & (x = \pm 2) \end{cases}$$

$$21. \text{ The } 2\text{-periodic function } g(x), \text{ where}$$

$$g(x) = x^2 \quad (-1 \leq x \leq 1)$$

$$23. \text{ The } 2\pi\text{-periodic function } g(x), \text{ where}$$

$$g(x) = \begin{cases} e^x & (-\pi < x < \pi), \\ (e^\pi + e^{-\pi})/2 & (x = \pm \pi) \end{cases}$$

$$25. \text{ (a) } F(x) = (x^2 - \pi^2)/2 \quad \text{(b) } F(x) = |x| - \pi$$

$$27. f(x) \sim \sum_{n=1}^{\infty} \frac{2}{(2n-1)\pi} \left[(-1)^{n+1} \cos \left(\frac{(2n-1)\pi x}{2} \right) + \sin \left(\frac{(2n-1)\pi x}{2} \right) \right]$$

$$29. c_0 = 0; \quad c_1 = 3/2; \quad c_2 = 0$$

Exercises 10.4, page 591

1. (a) The π -periodic function $\tilde{f}(x)$, where

$$\tilde{f}(x) = x^2 \quad (0 < x < \pi)$$
- (b) The 2π -periodic function $f_o(x)$, where

$$f_o(x) = \begin{cases} x^2 & (0 < x < \pi), \\ -x^2 & (-\pi < x < 0) \end{cases}$$
- (c) The 2π -periodic function $f_e(x)$, where

$$f_e(x) = \begin{cases} x^2 & (0 < x < \pi), \\ x^2 & (-\pi < x < 0) \end{cases}$$
3. (a) The π -periodic function $\tilde{f}(x)$, where

$$\tilde{f}(x) = \begin{cases} 0 & (0 < x < \pi/2), \\ 1 & (\pi/2 < x < \pi) \end{cases}$$

- (b) The
- 2π
- periodic function
- $f_o(x)$
- , where

$$f_o(x) = \begin{cases} -1 & (-\pi < x < -\pi/2), \\ 0 & (-\pi/2 < x < 0), \\ 0 & (0 < x < \pi/2), \\ 1 & (\pi/2 < x < \pi) \end{cases}$$

- (c) The
- 2π
- periodic function
- $f_e(x)$
- , where

$$f_e(x) = \begin{cases} 1 & (-\pi < x < -\pi/2), \\ 0 & (-\pi/2 < x < 0), \\ 0 & (0 < x < \pi/2), \\ 1 & (\pi/2 < x < \pi) \end{cases}$$

$$5. f(x) \sim -\frac{4}{\pi} \sum_{k=1}^{\infty} \frac{1}{2k-1} \sin(2k-1)\pi x$$

$$7. f(x) \sim \sum_{n=1}^{\infty} \left[\frac{2\pi(-1)^{n+1}}{n} + \frac{4}{\pi n^3} ((-1)^n - 1) \right] \sin nx$$

$$9. f(x) \sim \sum_{k=0}^{\infty} \frac{8}{(2k+1)^3 \pi^3} \sin(2k+1)\pi x$$

$$11. f(x) \sim \frac{\pi}{2} + \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} \cos(2k-1)x$$

$$13. f(x) \sim e - 1 + 2 \sum_{n=1}^{\infty} \frac{(-1)^n e - 1}{1 + \pi^2 n^2} \cos n\pi x$$

$$15. f(x) \sim \frac{2}{\pi} + \frac{2}{\pi} \sum_{k=1}^{\infty} \left(\frac{1}{2k+1} - \frac{1}{2k-1} \right) \cos 2kx$$

$$17. u(x, t) = \frac{2}{\pi} \sum_{k=1}^{\infty} \left[\frac{2}{2k-1} - \frac{1}{2k+1} - \frac{1}{2k-3} \right] e^{-5(2k-1)^2 t} \sin(2k-1)x$$

$$19. u(x, t) = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^{k+1}}{(2k+1)^2} e^{-5(2k+1)^2 t} \sin(2k+1)x$$

Exercises 10.5, page 603

$$1. u(x, t) = \sum_{n=1}^{\infty} \frac{8(-1)^{n+1} - 4}{\pi^3 n^3} e^{-5\pi^2 n^2 t} \sin n\pi x$$

$$3. u(x, t) = \frac{\pi}{2} - \sum_{k=0}^{\infty} \frac{4}{\pi(2k+1)^2} e^{-3(2k+1)^2 t} \cos(2k+1)x$$

$$5. u(x, t) = \frac{2(e^\pi - 1)}{\pi} + \sum_{n=1}^{\infty} \frac{2e^\pi(-1)^n - 2}{\pi(1+n^2)} e^{-n^2 t} \cos nx$$

$$7. u(x, t) = 5 + \frac{5}{\pi} x - \frac{30}{\pi} e^{-2t} \sin x + \frac{5}{\pi} e^{-8t} \sin 2x + \left(1 - \frac{10}{\pi} \right) e^{-18t} \sin 3x + \frac{5}{2\pi} e^{-32t} \sin 4x - \left(1 + \frac{6}{\pi} \right) e^{-50t} \sin 5x + \sum_{n=6}^{\infty} \frac{10}{\pi n} [2(-1)^n - 1] e^{-2n^2 t} \sin nx$$

$$9. u(x, t) = \frac{e^{-\pi} - 1}{\pi}x - e^{-x} + 1 + \sum_{n=1}^{\infty} c_n e^{-n^2 t} \sin nx,$$

where

$$c_n = \begin{cases} \frac{2e^{-\pi} - 2}{\pi n}(-1)^n + \frac{2n}{\pi(1+n^2)}[(-1)^{n+1}e^{-\pi} + 1] \\ + \frac{2}{\pi n}[(-1)^n - 1] & (n \neq 2), \\ \frac{e^{-\pi} - 1}{\pi} + \frac{4}{5\pi}(1 - e^{-\pi}) + 1 & (n = 2) \end{cases}$$

$$11. u(x, t) = \sum_{n=0}^{\infty} a_n e^{-4(n+1/2)^2 t} \cos(n+1/2)x, \text{ where}$$

$$f(x) = \sum_{n=0}^{\infty} a_n \cos(n+1/2)x$$

$$13. u(x, t) = \frac{\pi^2}{3}x - \frac{1}{3}x^3 - 3e^{-2t} \sin x \\ + \sum_{n=2}^{\infty} \frac{4(-1)^n}{n^3} e^{-2n^2 t} \sin nx$$

$$15. u(x, y, t) = e^{-52t} \cos 6x \sin 4y - 3e^{-122t} \cos x \sin 11y$$

$$17. u(x, y, t) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} e^{-n^2 t} \sin ny$$

$$19. C(x, t) = \sum_{n=1}^{\infty} c_n e^{-[L+kn^2\pi^2/a^2]t} \sin\left(\frac{n\pi x}{a}\right), \text{ where}$$

$$c_n = \frac{2}{a} \int_0^a f(x) \sin\left(\frac{n\pi x}{a}\right) dx$$

Concentration goes to zero as $t \rightarrow +\infty$.

Exercises 10.6, page 614

$$1. u(x, t) = \frac{1}{7\pi} \sin 7\pi t \sin 7\pi x \\ + \sum_{k=0}^{\infty} \frac{8}{((2k+1)\pi)^3} \cos(2k+1)\pi t \sin(2k+1)\pi x$$

$$3. u(x, t) = \sum_{n=1}^{\infty} \frac{4}{n^3} [2(-1)^{n+1} - 1] \cos 2nt \sin nx$$

$$5. u(x, t) = \frac{2h_0 L^2}{\pi^2 a(L-a)} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi a}{L} \sin \frac{n\pi x}{L} \cos \frac{n\pi \alpha t}{L}$$

$$7. u(x, t) = \cos t \sin x + \frac{5}{2} \sin 2t \sin 2x - \frac{3}{5} \sin 5t \sin 5x \\ + 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3} \left[t - \frac{\sin nt}{n} \right] \sin nx$$

$$9. u(x, t) = \sum_{n=0}^{\infty} \left[a_n \cos \frac{(2n+1)\pi \alpha t}{2L} \right. \\ \left. + b_n \sin \frac{(2n+1)\pi \alpha t}{2L} \right] \sin \frac{(2n+1)\pi x}{2L}, \text{ where}$$

$$f(x) = \sum_{n=0}^{\infty} a_n \sin \frac{(2n+1)\pi x}{2L} \text{ and}$$

$$g(x) = \sum_{n=0}^{\infty} b_n \frac{(2n+1)\pi \alpha}{2L} \sin \frac{(2n+1)\pi x}{2L}$$

$$11. u(x, t) = \sum_{n=1}^{\infty} a_n T_n(t) \sin \frac{n\pi x}{L}, \text{ where}$$

$$a_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx, \text{ and}$$

$$T_n(t) = e^{-t/2} \left(\cos \beta_n t + \frac{1}{2\beta_n} \sin \beta_n t \right), \text{ where}$$

$$\beta_n = \frac{1}{2L} \sqrt{3L^2 + 4\alpha^2 \pi^2 n^2}$$

$$13. u(x, t) = \frac{1}{2\alpha} [\sin(x + \alpha t) - \sin(x - \alpha t)] \\ = \frac{1}{\alpha} \sin \alpha t \cos x$$

$$15. u(x, t) = x + tx$$

$$17. u(x, t) = \frac{1}{2} \left[e^{-(x+\alpha t)^2} + e^{-(x-\alpha t)^2} \right. \\ \left. + \frac{\cos(x - \alpha t) - \cos(x + \alpha t)}{\alpha} \right]$$

$$21. u(r, t) = \sum_{n=1}^{\infty} [a_n \cos(k_n \alpha t) + b_n \sin(k_n \alpha t)] J_0(k_n r),$$

where

$$a_n = \frac{1}{c_n} \int_0^1 f(r) J_0(k_n r) r dr, \text{ and}$$

$$b_n = \frac{1}{\alpha k_n c_n} \int_0^1 g(r) J_0(k_n r) r dr, \text{ with}$$

$$c_n = \int_0^1 J_0^2(k_n r) r dr$$

Exercises 10.7, page 626

$$1. u(x, y) = \frac{4 \cos 6x \sinh[6(y-1)]}{\sinh(-6)} \\ + \frac{\cos 7x \sinh[7(y-1)]}{\sinh(-7)}$$

$$3. u(x, y) = \sum_{n=1}^{\infty} A_n \sin nx \sinh(ny - n\pi), \text{ where}$$

$$A_n = \frac{2}{\pi \sinh(-n\pi)} \int_0^{\pi} f(x) \sin nx dx$$

$$5. u(x, y) = \frac{\cos x \sinh(y-1)}{\sinh(-1)} - \frac{\cos 3x \sinh(3y-3)}{\sinh(-3)} \\ + \frac{\cos 2x \sinh 2y}{\sinh(2)}$$

$$7. u(r, \theta) = \frac{\pi}{2} - \sum_{k=0}^{\infty} \frac{r^{2k+1}}{(2k+1)^2 \pi 2^{2k-1}} \cos(2k+1)\theta$$

$$9. u(r, \theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{r}{a} \right)^n (a_n \cos n\theta + b_n \sin n\theta), \text{ where}$$

a_0 is arbitrary, and for $n = 1, 2, 3, \dots$

$$a_n = \frac{a}{\pi n} \int_{-\pi}^{\pi} f(\theta) \cos n\theta d\theta,$$

$$b_n = \frac{a}{\pi n} \int_{-\pi}^{\pi} f(\theta) \sin n\theta d\theta$$

$$11. u(r, \theta) = \left(\frac{1}{3}r - \frac{4}{3}r^{-1}\right)\cos\theta + \left(\frac{2}{3}r - \frac{2}{3}r^{-1}\right)\sin\theta \\ + \left(-\frac{1}{255}r^4 + \frac{256}{255}r^{-4}\right)\sin 4\theta$$

$$13. u(r, \theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} r^{-n}(a_n \cos n\theta + b_n \sin n\theta),$$

where

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\theta) \cos n\theta d\theta \quad (n = 0, 1, 2, \dots),$$

and

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\theta) \sin n\theta d\theta \quad (n = 1, 2, 3, \dots)$$

$$15. u(r, \theta) = r^3 \sin 3\theta$$

$$17. u(r, \theta) = a_0 + b_0 \ln r + \sum_{n=1}^{\infty} [(a_n r^n + b_n r^{-n}) \cos n\theta \\ + (c_n r^n + d_n r^{-n}) \sin n\theta],$$

where

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) d\theta, \quad b_0 = \frac{3}{2\pi} \int_{-\pi}^{\pi} g(\theta) d\theta,$$

and for $n = 1, 2, 3, \dots$

$$a_n + b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\theta) \cos n\theta d\theta,$$

$$n3^{n-1}a_n - n3^{-n-1}b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} g(\theta) \cos n\theta d\theta,$$

and

$$c_n + d_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\theta) \sin n\theta d\theta,$$

$$n3^{n-1}c_n - n3^{-n-1}d_n = \frac{1}{\pi} \int_{-\pi}^{\pi} g(\theta) \sin n\theta d\theta$$

$$21. u(r, \theta, z) = [I_0(r)/I_0(\pi)] \sin z$$

$$23. (c) \phi(x, y) = 2xy$$

APPENDIX A

Exercises A, page A-1

$$1. (x^2 + 1)^{5/2}(5x^2 - 2)/35 + C \quad 3. \frac{2}{15}e^{6r^5} + C$$

$$5. \frac{1}{2}\theta + \frac{1}{4\pi}\sin(2\pi\theta) + C$$

$$7. \frac{1}{2}\ln(t^2 + 4) + \frac{1}{2}\arctan(t/2) + C$$

$$9. (1/\sqrt{2}) \arcsin \sqrt{2}x + C$$

$$11. 2\ln|x-1| - 4\ln|x-2| + 2\ln|x-3| + C$$

$$13. \sqrt{9x^2 - 1} - \arccos\left(\frac{1}{3x}\right) + C$$

$$15. \frac{1}{2}\ln(x^2 + 4) - \frac{2}{x} + C \quad 17. \ln|\ln x| + C$$

$$19. y \cosh y - \sinh y + C \quad 21. \frac{x}{4\sqrt{x^2 + 4}} + C$$

$$23. \frac{1}{2}(t^2 - 9)\ln(t + 3) - \frac{t^2}{4} + \frac{3t}{2} + C$$

$$25. -\arctan(\cos t) + C$$

$$27. \frac{5}{7}\sin^{7/5}x - \frac{5}{17}\sin^{17/5}x + C$$

$$29. \frac{\sin(10x)}{20} + \frac{\sin(4x)}{8} + C$$

$$31. \frac{1}{9}\tan^9\theta + \frac{2}{7}\tan^7\theta + \frac{1}{5}\tan^5\theta + C$$

$$33. \frac{1}{8}x - \frac{1}{96}\sin(12x) + C$$